VIGNAN's

INSTITUTE OF ENGINEERING FOR WOMEN

(AUTONOMOUS)

(Approved by AICTE & Affiliated to JNTU-GV)
Accredited by NAAC with 'A+' Grade, NBA for UG Programmes of EEE, ECE, CSE & IT
Kapujaggarajupeta, Visakhapatnam

DEPARTMENT OF

BASIC SCIENCES AND HUMANITIES (BS&H)

ADVANCED CHEMISTRY LAB MANUAL & RECORD



Name: _____

Year: _____ Sem: _____ Regd No: _____

Academic Year: _____

VIGNAN'S INSTITUTE OF ENGINEERING FOR WOMEN (A)



ADVANCED CHEMISTRY LABORATORY

MANUAL & RECORD

(VR-26 FOR ALL PROGRAMMES)

CERTIFICATE

This is to certify that **Ms.** _____, bearing **Roll No.** _____, a student of **B.Tech.** _____**Year** _____ **Semester**, **Department of** _____, at VIGNAN'S INSTITUTE OF ENGINEERING FOR WOMEN (A):has satisfactorily completed the experiments prescribed in the **Chemistry Laboratory** during the **Academic Year** ____.

Laboratory In – Charge **Head of the Department**

ADVANCED CHEMISTRY LAB

LAB SAFETY

SAFETY FIRST – PREVENT ACCIDENTS

PERSONAL PROTECTIVE EQUIPMENT (PPE)



- **LAB COOT** — Wear clean lab coat
- **SAFETY GOGGLES** — Protect your eyes
- **GLOVES** — Wear suilable gloves
- **CLOSED SHOES** — Wear closed iiable at all times

GENERAL LAB SAFETY RULES

- Read the experiment and instructions carefully before starting.
- Do not eat, drink, smoks or shew gum in the laboratory.
- Wear appropriate PPE at all times.
- Do not perform inwathorized experiments.
- Keep your work area clean and organized.
- Do not taste any chemicals. Avoid smelling chemicals directly.
- Report any accident, spill or breakage immediately.
- Wash your hands with soap after completing the experiment.
- Never work alone in the laboratory.

CHEMICAL SAFETY

- Read labals on all chemical bottles before use.
- Use only the required amount of chemicals.
- Do not return anused chemicals to the stock bottle.
- Keep chemicals in proper labeled containers.
- Store acids and bases separately.
- Handie corrosive and toxic chemicals carefully.



Corrosive · Toxic · Oxidizing



Flammable · Irritant · Environmental Hazard

IN CASE OF EMERGENCY

- In case of fire, switch off gas and electricity. Use fhe extinguisher.
- For chemical spills on skin or eyes, wash immediately with plenty of water and inform the instructor.
- For eye contact, wash eyes with water for at least 15 minutes and usek medical help.
- Report all accidents immediately.

FIRE SAFETY

- Keep flammable chemicals away from heat and open tiames.
- Do not leave experiments unattended.
- Know the location of the eatinguisner, sand bucket, fire bianket and first aid box.



HANDLING OF LABORATORY GLASSWARE

GENERAL PRECAUTIONS FOR GLASSWARE

- Inspect glassware before use.
- Do not use cracked or chipped glassware.
- Handle glassware with care.
- Do not apply excessive force.
- Use appropriate glassware for the experiment.
- Clean glassware immediately after use.
- Keep glassware away from the edge of the taste.
- Do not use glassware for purpose other than intended.
- Place hat glassware on wire gaoze or asbestos sheet.
- Store glassware in designated places.

CLEANING OF GLASSWARE

- Rinse with tap water to remove chemical resiituss.
- Wash with suitable detergent using a brush.
- Aloxe thoroughly with tap water followed by distilled water.
- Allow to dry in an intersed position.



TYPES OF GLASSWARE AND HOW TO HANDLE

BEAKER



- For mixing and heating beaskers.
- Do not heat aooply beakers.
- Ilio ewin gaon side beakers.

CONICAL FLASK



- For treations and mising
- Swilt gwidly, do not shake sigarevisly.
- Do not heat closed conical 6asks.

VOLUMETRIC PLASK



- For preparing standard solutions.
- Fill up io the mank unit;
- Do not heat.
- Stopper with care.

MEASURING CYLINDER



- For measuring approximate solutions.
- Read the maniscus at nse hoat.
- Do not heat.

BURETTE



- For accurate delivery of solutions.
- Clamp wertwolly.
- Fill without ait bubbiws.
- Read the menissus at eye level.
- Do not force the stoppock.

PIPETTE (VOLUMETRIC)



- For accurate trarater of fixed rehsing.
- Use pipette tule; never use meuth.
- Do not heat.
- Rinse before use.

FUNNEL



- For filtration and transferring hawds.
- Keep the sten touching the wall of the receiving fiask.
- Use fiter paper properly.

WATCH CLASS



- For covering beakers or evaporating small amounts.
- Handle with loncepe when hot.

TEST TUBE



- For small scale reactions.
- Hald with a toer take holder while heating.
- Keep this meuth away from people.

ROUND BOTTOM FLASK



- For heating and refioxing.
- Use pipate ress or wire gasse.
- Do not hesk empty.

SEPARATING FUNNEL



- For liquid -liquid extraction.
- Remove the stapper before opening the stoppock.
- Hold carefully.

DESICCATOR



- For drying and cooling somples.
- Grease the juint dightly.
- Do net force fice isk

DO'S

- Always keep your work area clean and dry.
- Use appropriate glassware for the task.
- Handle glassware gradty.
- Report breakage immediately.
- Dispose broken glass in designated contsieat.

DON'TS

- Do not use cracked or chipped glassware.
- Do not heat closed glassware.
- Do not use excessive force while cleaning.
- Do not place hot glassware on soid surface.
- Do not clutter the werk bench.

BROKEN GLASSWARE



- Do not pick up broken glass with bare bands.
  Use brush and dostpon.
- Dispose in the broken glass waste container.
  Inform the instructor.

★ SAFETY IS NOT EXPENSIVE, IT IS PRICELESS. THINK SAFETY, WORK SAFELY, BE SAFE. ★

INDEX

Sno	Date	Name of the Experiment	Marks	Signature
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

No. of Experiments done: _____

Avg marks_____

PREPARATION OF STANDARD SOLUTIONS

1 Introduction

Preparation of standard solutions is one of the most fundamental operations in analytical chemistry. A standard solution is a solution whose concentration is accurately known and is used for quantitative chemical analysis, standardization of other solutions, and volumetric titrations. The accuracy of any analytical experiment largely depends on the proper preparation and standardization of the reagents used.

In the Chemistry Laboratory, standard solutions are extensively employed in titrimetric, conductometric, and water quality analyses. Careful weighing of chemicals, accurate measurement of volumes, proper dissolution, and storage of solutions are essential for obtaining reliable and reproducible results.

2 Objectives

After completing this exercise, the student will be able to:

- Understand the concept of standard solutions.
- Differentiate between primary and secondary standards.
- Prepare solutions of required molarity and normality.
- Perform accurate weighing and dilution techniques.
- Handle volumetric glassware properly.
- Store and label laboratory reagents safely.

3 Types of Standard Solutions

Primary Standard Solution

A primary standard solution is prepared by dissolving an accurately weighed quantity of a highly pure, stable, non-hygroscopic substance in a known volume of distilled water.

Characteristics

- High purity (≥99.9%)
- Stable in air
- Non-hygroscopic
- High molecular weight
- Easily soluble in water

- Does not react with atmospheric gases

Examples

- Sodium carbonate (Na_2CO_3)
- Oxalic acid ($H_2C_2O_4 \cdot 2H_2O$)
- Potassium dichromate ($K_2Cr_2O_7$)
- Potassium hydrogen phthalate (KHP)

Secondary Standard Solution

Secondary standard solutions cannot be prepared accurately by direct weighing because they are unstable or hygroscopic. Therefore, they are standardized against a primary standard.

Examples

- Hydrochloric acid (HCl)
- Sodium hydroxide (NaOH)
- Potassium permanganate ($KMnO_4$)
- Silver nitrate ($AgNO_3$)
- EDTA

4 Laboratory Glassware Required

- Volumetric Flask (100, 250, 500 and 1000 mL)
- Pipette
- Burette
- Measuring Cylinder
- Funnel
- Beaker
- Glass Rod
- Wash Bottle
- Analytical Balance

5 General Procedure for Preparing Standard Solutions

1. Calculate the required quantity of solute.
2. Accurately weigh the chemical using an analytical balance.
3. Transfer the chemical into a clean beaker.
4. Dissolve it in a small quantity of distilled water.
5. Transfer the solution into a volumetric flask using a funnel.

6. Wash the beaker several times and transfer the washings into the flask.
7. Make up the solution exactly to the calibration mark using distilled water.
8. Stopper and mix thoroughly.
9. Label the bottle with:
    - Name of solution
    - Concentration
    - Date of preparation
    - Prepared by

6 Formulae Used

Molarity

$$\boldsymbol{M = \frac{weight}{gram\ molecular\ weight} \times \frac{1000}{volume\ in\ ml}}$$

Normality

$$\boldsymbol{N = \frac{weight}{gram\ equivalent\ weight} \times \frac{1000}{volume\ in\ ml}}$$

Dilution Formula

$$N_1V_1 = N_2V_2$$

LIST OF EXPERIMENTS

1. **Determination of the band-gap energy of a semiconductor sample from its optical absorption edge.**
2. **Conductometric titration of a strong acid against a strong base.**
3. **Determination of the EMF of a Daniel cell and verification of the Nernst equation.**
4. **Determination of the charge–discharge characteristics of a lithium-ion battery / supercapacitor cell.**
5. **Determination of the specific and equivalent conductance of an electrolyte solution.**
6. **Determination of the corrosion rate of a copper / mild-steel specimen by the weight-loss method.**
7. **Preparation and observation of colloidal nanoparticles (size-dependent colour-change demonstration).**
8. **Determination of the pH and conductivity of an electronic-grade (deionized) water sample used in wafer/PCB cleaning.**
9. **Potentiometry - determination of redox potentials and emfs**
10. **Determination of total hardness of a water sample by the EDTA titrimetric method.**
11. **Determination of acidity and alkalinity of a water sample by acid titration.**
12. **Determination of dissolved oxygen (DO) in a water sample by Winkler's method**
13. **Preparation and study of phenol-formaldehyde resin sample.**

EXPERIMENT – 1

ESTIMATION OF TOTAL HARDNESS OF WATER BY EDTA METHOD (COMPLEXOMETRIC TITRATION)

Aim

To estimate the total hardness, permanent hardness and temporary hardness of water by using standard solution of EDTA

Apparatus Required

- Burette (50 mL)
- Pipette (20 mL)
- Conical flask (250 mL)
- Volumetric flask (250 mL)
- Measuring cylinder (50 mL)
- Funnel
- Beaker
- Glass rod
- Wash bottle
- White tile
- Burette stand
- Hot plate

Chemicals Required

- Standard EDTA solution (0.01 N)
- Ammonia-Ammonium Chloride ($NH4Cl+NH_4OH$) Buffer Solution (pH 10)
- Eriochrome Black-T (EBT) Indicator
- Distilled Water
- Given Water Sample
- Standard MgSO4

Principle

EDTA is a well-known complexing agent, forms stable complex with Ca and Mg ions as well as many other metal ions in aqueous solution. It reacts with metal ions 1:1 ratio.

Hard water which contains Ca^{2+} and Mg^{2+} ions react with EBT forms unstable wine-red color complex at pH 9-10.

$$Ca^{2+} \text{ (or) } Mg^{2+} + EBT \rightarrow [Ca\text{-}EBT] \text{ (or) } [Mg\text{-}EBT]$$

(Wine red color complex)

During the titration, the wine red-colored complexes [Ca-EBT] (or) [Mg-EBT] react with EDTA and forms a stable colorless complex [Ca-EDTA] (or) [Mg-EDTA] by releasing blue colour indicator into the solution. Hence the color change at end point is wine red to blue.

[Ca-EBT] (or) [Mg-EBT] + EDTA→ [Ca-EDTA] (or) [Mg-EDTA] + EBT

(Wine red color complex) (Colorless stable complex) (Blue)

Procedure:

Step-1: Determination of Total Hardness:

- Take 50ml of water sample into a conical flask add 2 ml of buffer solution add 2- 3drops of EBT indicator.
- Titrate the wine-red color solution with EDTA taken in burette, till a blue color end point is obtained.
- Repeat the titration to get concurrent values.

End Point: Wine red color solution changes to pale blue

Observations:

S. No	**Volume of Hard water in (ml) (V_1)**	**Burette Reading**		**Volume of EDTA consumed(V_2) ml**
		Initial	**Final**	
1	50.0			
2	50.0			
3	50.0			

Concordant volume of EDTA (V_2) = ml

M1= Molarity of water sample =

V1 = volume of water sample = 50ml

M2= EDTA Molarity =

V2= volume of EDTA consumed = ml

$$M_1V_1/n_1 = M_2V_2/n_2; \quad n_1=n_2=1$$

$$M_1= M_2V_2/V_1$$

M_1 = Molarity of water sample = M

Total hardness = $M_1 \times 100 \times 1000$ = ppm

RESULT: The hardness of the given water sample has been found to be as follows:

Total hardness = ppm

Step-2: Determination of Permanent Hardness:

- Take 250ml of sample hard water in a large beaker.
- Boil it, till the volume reduced to about 50ml.
- Filter, Wash the Precipitate with distilled water, collecting filtrate and washings in a 250 ml measuring flask.
- Finally, make up the volume to 250ml with distilled water.
- Take 50ml of water sample into a conical flask add 2 ml of buffer solution add 2- 3drops of EBT indicator.
- Titrate the wine-red color solution with EDTA taken in burette, till a blue color end point is obtained.
- Repeat the titration to get concurrent values.

End Point: Wine red color solution changes to pale blue

OBSERVATIONS:

S. No	**Volume of Hard water in (ml) (V_3)**	**Burette Reading**		**Volume of EDTA consumed(V_4) ml**
		Initial	**Final**	
1	50.0			
2	50.0			
3	50.0			

Concordant volume of EDTA (V_4) = ml

M_3= Molarity of water sample =M

V_3 = volume of water sample = 50ml

M_4= EDTA Molarity = M

V_4= volume of EDTA consumed = ml

$$M_3V_3 = M_4V_4$$

$$M_3 = M_4V_4/V_3$$

M_3 = Molarity of boiled hard water sample = M

Permanent hardness = $M_3 \times 100 \times 1000$ = ppm

Total hardness= Permanent hardness+ Temporary hardness

Temporary hardness= Total hardness- Permanent hardness

=.....ppm

RESULT:

The hardness of the given water sample has been found to be as follows:

Total hardness =_____ppm

Permanent hardness=_____ppm

Temporary hardness=__ppm

VIVA QUESTIONS

1. What is hardness of water?

Hardness is the property of water caused mainly by dissolved **Ca^{2+} and Mg^{2+} salts**, which prevents water from forming lather readily with soap.

2. What are the two types of hardness?

Temporary hardness & Permanent hardness

3. What causes temporary hardness?

Temporary hardness is mainly caused by **bicarbonates of calcium and magnesium**, such as $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$.

4. What causes permanent hardness?

Permanent hardness is mainly due to **chlorides and sulphates of calcium and magnesium**, such as $CaCl_2$, $MgCl_2$, $CaSO_4$ and $MgSO_4$.

5. Which method is used for estimation of hardness in this experiment?

The **complexometric EDTA titration method** is used.

6. What is the full form of EDTA?

EDTA stands for **Ethylenediaminetetraacetic acid**.

7. Why is EDTA used?

EDTA forms stable complexes with metal ions such as **Ca^{2+} and Mg^{2+}** in a 1:1 ratio.

8. What is the indicator used?

Eriochrome Black-T (EBT) is used as the indicator.

9. What is the colour change at the end point?

The colour changes from **wine red to blue**.

10. Why is buffer solution added?

The buffer maintains the solution at approximately **pH 10**, which is the optimum pH for the EDTA titration using EBT.

11. Why is hardness expressed as $CaCO_3$?

$CaCO_3$ is used as the standard reference substance because its molecular/equivalent mass provides a convenient basis for expressing different hardness-producing salts on a common scale.

12. What is the unit of hardness?

Hardness is commonly expressed as **mg/L or ppm as $CaCO_3$**.

13. What is the relationship between mg/L and ppm for dilute water solutions?

For dilute aqueous solutions:

1 mg/L ≈ 1 ppm

14. What is the role of EBT?

EBT acts as a **metallochromic indicator** and forms a coloured complex with Ca^{2+}/Mg^{2+} ions.

15. What happens at the end point?

EDTA complexes the metal ions completely, displacing EBT from the metal–indicator complex.

The free indicator produces the **blue colour**.

16. Why is EDTA titration called a complexometric titration?

It is called complexometric titration because the titration is based on the formation of a **stable metal–EDTA complex**.

17. What is the ratio of metal ion to EDTA in the complex?

The metal ion and EDTA generally react in a **1:1 molar ratio**.

18. What is the difference between hard water and soft water?

Hard water contains appreciable amounts of Ca^{2+} and Mg^{2+} ions and does not readily form lather with soap. Soft water contains very low concentrations of these ions and forms lather easily.

19. What is meant by total hardness?

Total hardness is the combined hardness due to **calcium and magnesium ions**, including both temporary and permanent hardness.

20. What is the end point in this experiment?

The end point is reached when the solution changes from **wine red to blue**.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 2

ESTIMATION OF ACIDITY & ALKALINITY OF WATER SAMPLE

AIM

To determine the acidity of the given water sample by acid -base titration using standard sodium hydroxide.

To determine the **phenolphthalein alkalinity (P-alkalinity)** and **total alkalinity (T-alkalinity)** of the given water sample by acid-base titration using standard hydrochloric acid.

Apparatus Required

- Burette (50 mL)
- Pipette (20 mL)
- Conical flask (250 mL)
- Volumetric flask
- Measuring cylinder
- Funnel
- Beaker
- Glass rod
- Wash bottle
- White tile

Chemicals Required

- Standard Sodium hydroxide (0.02 N)
- Standard Hydrochloric Acid (0.02 N)
- Phenolphthalein Indicator
- Methyl Orange Indicator
- Distilled Water
- Given Water Sample

Theory

The acidity of water represents its ability to give protons that comefrom:1. De-ionized molecules of ionized weak acids such as (carbonic acid and tannic acid).2. Ferrous and aluminum salts3. Weak mineral acids. Such as (sulfuric acid or hydrochloric acid).CO2 is the most likely cause of acidity in water, its result of respiration and autolysis Process of plants & animals in water. CO2 concentrations effect to (pH) values in water that have different effects on aquatic organism, some of organism can lived in acidic medium reach to (pH = 2). So pH values vary throughout the day due to respiration and photosynthesis process that cased different acidity values in water. Alkalinity is the capacity of water to neutralize acids and is mainly due to the presence of hydroxide (OH^-), carbonate (CO_3^{2-}), and bicarbonate (HCO_3^-) ions. It is an important

parameter in water quality analysis because it indicates the buffering capacity of water and its ability to resist sudden changes in pH.

The alkalinity of natural waters is generally caused by dissolved carbonates and bicarbonates of calcium, magnesium, sodium, and potassium. Excessive alkalinity may impart an unpleasant taste, cause scale formation, interfere with industrial processes, and affect water treatment operations.

Acidity is expressed as mg/L

Alkalinity is expressed as **mg/L (ppm) as $CaCO_3$ equivalent**.

Principle

The acidity of water is determined by titrating with water sample with a standard sodium hydroxide solution using phenolphthalein indicator.

The alkalinity of water is determined by titrating the water sample with a standard hydrochloric acid solution using two indicators.

- **Phenolphthalein** indicates the neutralization of hydroxide ions and half of the carbonate ions. The end point is indicated by the disappearance of the pink colour (pH ≈ 8.3). The volume of acid required gives the **Phenolphthalein Alkalinity (P-Alkalinity)**.
- **Methyl Orange** indicates the complete neutralization of hydroxide, carbonate, and bicarbonate ions (pH ≈ 4.5). The end point is indicated by the colour change from yellow to orange/red. The total volume of acid consumed gives the **Total Alkalinity (T-Alkalinity)**.

Chemical Reactions

Neutralization of Hydroxide

$$OH^- + H^+ \rightarrow H_2O$$

Neutralization of Carbonate

$$CO^{2-} + H^+ \rightarrow HCO_3^-$$

Neutralization of Bicarbonate

$$HCO_3^- + H^+ \rightarrow H_2CO_3$$

$$H_2CO_3 \rightarrow CO_2 + H_2O$$

Procedure

Determination of acidity

1. Take 100 ml from water sample either supplement or irrigated water by cylinderand put it in a flask.

2. Add 3 drops from phenolphthalein as indicator solution.

3. Titrate with 0.025N sodium hydroxide solution (NaoH).

4. Stir the water sample gently during the titration.

5. The (end point) of titration is the start of pink color appearance in the solution.

6. Record the volume of (NaoH) and calculate water acidity by following equation:

Acidity = $(\boldsymbol{ml\ NaOH\ titrant})*(\boldsymbol{Normality\ NaOH})*\mathbf{1000}/(\boldsymbol{ml\ water\ sample})$

A. Determination of Phenolphthalein Alkalinity (P-Alkalinity)

1. Rinse the burette with **0.02 N HCl** and fill it.
2. Pipette **20 mL** of the given water sample into a clean conical flask.
3. Add **2–3 drops** of phenolphthalein indicator.
4. If the solution remains colourless, the **P-alkalinity is zero**.
5. If a pink colour develops, titrate with standard HCl until the pink colour just disappears.
6. Record the burette reading as **P mL**.

B. Determination of Total Alkalinity (T-Alkalinity)

1. Take another **20 mL** of the water sample in a clean conical flask.
2. Add **2–3 drops** of methyl orange indicator.
3. Titrate against **0.02 N HCl** until the colour changes from **yellow to orange/red**.
4. Record the burette reading as **T mL**.
5. Repeat the titration to obtain concordant readings.

Observation Table

Acidity:

Burette Solution

0.02 N Sodium hydroxide

Pipette Solution

100 ml Water Sample

Indicator

Phenolphthalein

Trial No.	Initial Reading (mL)	Final Reading (mL)	Base Used (mL)
Rough			
I			
II			
III			

Alkalinity:

Burette Solution

0.02 N Hydrochloric Acid

Pipette Solution

20 mL Water Sample

Indicator

Phenolphthalein

Trial No.	Initial Reading (mL)	Final Reading (mL)	Acid Used (P mL)
Rough			
I			
II			
III			

Average Phenolphthalein Reading = _____mL

Indicator

Methyl Orange

Trial No.	Initial Reading (mL)	Final Reading (mL)	Acid Used (T mL)
Rough			
I			
II			
III			

Average Total Alkalinity Reading = _____mL

Calculations

Acidity General Formula

$$[N_1V_1 = N_2V_2]$$

Where:

- $\mathbf{N_1}$ = Normality of NaOH
- $\mathbf{V_1}$ = Volume of NaOH used (mL)
- $\mathbf{N_2}$ = Normality of alkalinity

- **V_2** = Volume of water sample (mL)

Acidity

$$= \frac{V_1 \times N_1 \times 50 \times 1000}{V_2}$$

Alkalinity General Formula

$$[N_1V_1 = N_2V_2]$$

Where:

- **N_1** = Normality of HCl
- **V_1** = Volume of HCl used (mL)
- **N_2** = Normality of alkalinity
- **V_2** = Volume of water sample (mL)

Phenolphthalein Alkalinity

$$= \frac{P \times N \times 50 \times 1000}{V_S}$$

Where:

P = Volume of acid required for phenolphthalein end point (mL)

- **N** = Normality of HCl
- **50** = Equivalent weight of $CaCO_3$
- **V_S** = Volume of sample (20 mL)

Total Alkalinity

$$= \frac{T \times N \times 50 \times 1000}{V_S}$$

Where:

T = Volume of acid required for methyl orange end point (mL)

Sample Calculation

Given:

- Average acid used (T) = **12.5 mL**
- Normality of HCl = **0.02 N**
- Sample volume = **20 mL**

$$\text{Total Alkalinity (as } CaCO_3) = \frac{12.5 \times 0.02 \times 50 \times 1000}{20}$$

= 625 mg/L

Result

Acidity = _____ mg/L (as $CaCO_3$)

Phenolphthalein Alkalinity = **_____mg/L (as $CaCO_3$)**

Total Alkalinity = **_____mg/L (as $CaCO_3$)**

Precautions

1. Use freshly prepared standard hydrochloric acid.
2. Use clean and dry glassware.
3. Add only **2–3 drops** of indicator.
4. Observe the end point carefully against a white background.
5. Perform the titration slowly near the end point.
6. Avoid overshooting the end point.
7. Record burette readings at eye level.
8. Obtain concordant readings for accurate results.
9. Mix the solution thoroughly during titration.
10. Wash the burette with HCl solution before filling.

Applications

- Drinking water quality analysis.
- Water treatment plant operation.
- Boiler feed water analysis.
- Industrial process water monitoring.
- Environmental water quality assessment.
- Wastewater treatment studies.
- Determination of the buffering capacity of natural waters.

Viva Questions

1. What is alkalinity?

   The capacity of water to neutralize acids, mainly due to bicarbonates, carbonates, and hydroxides.
2. Which ions contribute to alkalinity?

   OH^-, CO_3^{2-}, and HCO_3^- ions.
3. Why is alkalinity expressed as $CaCO_3$?

   Because $CaCO_3$ provides a convenient standard reference with a molecular equivalent weight of 50 g/eq.
4. Why are two indicators used in this experiment?

   They are used to identify different neutralization end points and determine phenolphthalein and total alkalinity.

5. What is the end point of phenolphthalein alkalinity?

   The colour changes from pink to colourless, corresponding approximately to pH 8.3.
6. What is the end point of methyl orange alkalinity?

   The colour changes from yellow to orange/red, corresponding approximately to pH 4.5.
7. What is the significance of total alkalinity?

   It indicates the total acid-neutralizing capacity of water and its buffering ability.
8. What is the difference between alkalinity and pH?

   pH indicates the intensity of acidity/alkalinity, whereas alkalinity indicates the capacity to neutralize acid.
9. Why is HCl used as the titrant?

   HCl is a strong acid that reacts quantitatively with alkaline ions in water.
10. Mention two applications of alkalinity determination.

    Alkalinity determination is used in water-treatment control and boiler/wastewater management.

Report

Title

Estimation of Alkalinity of Water Sample

Experimental Data

- Volume of sample taken = _____mL
- Normality of HCl = _____N
- Average P-alkalinity titre = _____mL
- Average T-alkalinity titre = _____mL

Calculations

Result

Acidity = _____ mg/L (as $CaCO_3$)

Phenolphthalein Alkalinity = _____**mg/L (as $CaCO_3$)**

Total Alkalinity = _____**mg/L (as $CaCO_3$)**

Interpretation

- **P = 0:** Only bicarbonate alkalinity is present.
- **P < ½T:** Carbonate and bicarbonate alkalinity are present.

- **P = ½T:** Only carbonate alkalinity is present.
- **P > ½T:** Hydroxide and carbonate alkalinity are present.
- **P = T:** Only hydroxide alkalinity is present.

Conclusion

The acidity of the given water sample was successfully determined by acid-base titration using standard sodium hydroxide with phenolphthalein indicator. The results indicate the buffering capacity and the nature of acid constituents present in the water sample. The alkalinity of the given water sample was successfully determined by acid-base titration using standard hydrochloric acid with phenolphthalein and methyl orange indicators. The results indicate the buffering capacity and the nature of alkaline constituents present in the water sample.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 3

ESTIMATION OF DISSOLVED OXYGEN (DO) IN WATER BY WINKLER'S METHOD

AIM: To Determine Dissolved Oxygen in a Sample of Water.

APPARATUS: Burette, pipette, conical flask, beaker, measuring cylinder

CHEMICALS: $MnSO_4$, KOH, KI, $Na_2S_2O_3$, NaN_3, Starch, H_2SO_4, alkali-iodide-azide

Theory: It is based on oxidation of potassium iodide. The liberated iodine is titrated against standard hypo solution using starch as a final indicator. Since oxygen in water is in molecular state and not capable to react with KI, an oxygen carrier manganese hydroxide is used to bring about the reaction between KI and O2. Manganous hydroxide is produced by the action of potassium hydroxide and manganese sulphate.

Chemical reaction:

$$2KOH+MnSO_4 \rightarrow Mn(OH)_2+K_2SO_4$$

$$2Mn(OH)_2+O_2 \rightarrow 2MnO(OH)_2$$

$$MnO(OH)_2+H_2SO_4 \rightarrow MnSO_4+2H_2O+[O]$$

$$2KI+H_2SO_4+[O] \rightarrow K2SO4+H_2O+I_2$$

$$I_2+2Na_2S_2O_3 \rightarrow 2NaI+Na_2S_4O_6$$

Procedure:

1. Take 500ml of water in a D.O bottle.
2. Add 10ml of alkali-iodide-azide and 10 ml of MnSO4 into it.
3. Stopper the bottle and shake it well.
4. Keep the bottle in dark for 5 min and add conc. H2SO4 till the brown precipitates are dissolved.
5. Take 100 ml of the above solution in a conical flask.
6. Titrate against hypo till the color changes to light Yellow.
7. Add 3-4 drops of starch in to it and the color changes to blue.
8. The blue color solution is titrated against hypo solution till blue color disappear
9. This is end point of the titration. Repeat this process till to get three concordant reading.

Observation:

S.No.	Volume of water sample solution taken in ml.(V1)	Burette readings		Volume of hypo Solution consumed in ml.**(V2)**
		Initial	Final	
1				
2				
3				

Calculation:

Volume of water (V1) = ml

Normality of water (N1) = N

Volume of hypo (V2) =ml

Normality of hypo ($N_{2)}$ = N

By volumetric principle $V_1 N_1 = V_2 N_2$

Normality of water (N1) = N

Amount of DO present in given water sample solution= $N_1 \times 8 \times 1000$ ppm

Report: The amount of dissolved oxygen in a sample of water is found to be….. ppm.

VIVA QUESTIONS

1. What is dissolved oxygen?

Dissolved oxygen is the amount of molecular oxygen (**O_2**) dissolved in water.

2. In which units is DO expressed?

DO is generally expressed in **mg/L (milligrams per litre)** or ppm.

3. Which method is used to estimate DO?

The **Winkler's iodometric method** is commonly used.

4. What is the principle of Winkler's method?

Dissolved oxygen oxidizes a manganous compound, which on acidification liberates iodine from iodide. The liberated iodine is titrated with sodium thiosulphate.

5. Why is manganous sulphate used?

Manganous sulphate provides **Mn^{2+} ions**, which react with the alkaline reagent and subsequently participate in the reaction with dissolved oxygen.

6. Why is alkaline KI reagent added?

It provides an alkaline medium and **iodide ions**, which ultimately allow iodine to be liberated in an amount equivalent to the dissolved oxygen.

7. Why is sodium azide present in the reagent?

Sodium azide suppresses interference from **nitrite ions**, which can otherwise oxidize iodide and produce an erroneous DO value.

8. Why is sulphuric acid added?

Sulphuric acid dissolves the manganese precipitate and acidifies the solution, allowing iodine to be liberated from iodide.

9. Which indicator is used?

Starch is used as the indicator.

10. What is the colour change at the end point?

The solution changes from **blue to colourless**.

11. Why is starch added near the end point?

Adding starch near the end point gives a sharper end point and avoids formation of a very stable starch–iodine complex at high iodine concentrations.

12. What is the role of iodine in this experiment?

Iodine acts as the intermediate species whose amount is equivalent to the dissolved oxygen present in the original water sample.

13. Why should air bubbles be avoided?

Air contains oxygen. If an air bubble enters the sample, oxygen can dissolve into the water and cause the measured DO to be **higher than the actual value**.

14. Why is the sample fixed immediately?

DO can change due to contact with air, microbial activity, temperature changes, and chemical

reactions. Immediate fixation preserves the oxygen content at the time of sampling.

15. What happens if the water sample contains more dissolved oxygen?

A greater amount of iodine is liberated, requiring a greater volume of sodium thiosulphate for titration.

16. What is the function of sodium thiosulphate?

It reduces iodine to iodide and is used to determine the amount of iodine liberated.

17. What is the significance of DO in water?

DO is an important indicator of **water quality** and the ability of water to support aquatic life.

18. What happens to DO when organic pollution increases?

Generally, DO decreases because microorganisms consume oxygen while decomposing organic matter.

19. What is the approximate DO of clean, oxygenated water?

It depends strongly on temperature, pressure, salinity, and other conditions. At ordinary temperatures, clean freshwater often contains several mg/L of dissolved oxygen, commonly around **8–9 mg/L near room temperature when air-saturated**.

20. What is the relationship between DO and BOD?

BOD (Biochemical Oxygen Demand) measures the oxygen consumed by microorganisms while decomposing biodegradable organic matter, whereas **DO** is the oxygen actually dissolved in the water.

21. What does low DO indicate?

Low DO may indicate **organic pollution, microbial activity, stagnation, or poor aeration**.

22. Why is the water sample taken in a BOD bottle?

A BOD bottle is designed to minimize contact between the sample and atmospheric air, helping preserve the original dissolved oxygen concentration.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 4

CONDUCTOMETRIC TITRATION: STRONG ACID VS STRONG BASE

AIM

To determine the **strength/concentration of a strong acid (e.g., HCl)** by titrating it against a **standard solution of a strong base (e.g., NaOH)** using the **conductometric method**.

Apparatus Required

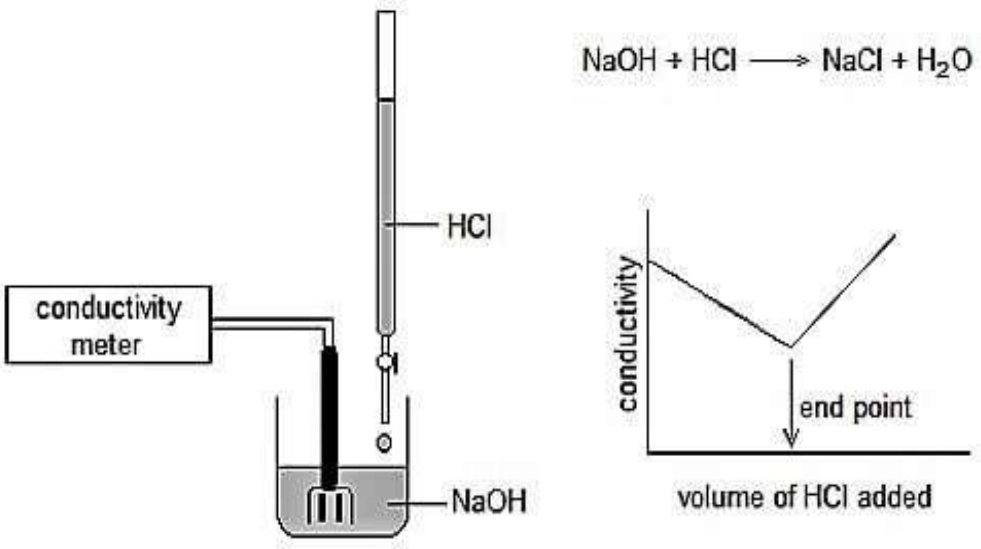
- Conductivity Cell-1
- Burette (50 mL)
- Conductivity Meter-1
- Glass beaker – 250ml-1
- Glass beaker-100ml-1
- Glass Rod-1

Chemicals Required

- 1.0N standard Sodium Hydroxide(NAOH) -100ml
- 0.1 N Hydrochloric acid (HCL) -100ml
- Distilled water

Principle:

The conductance of the solution changes as the strong acid reacts with the strong base. The **equivalence point** is determined from the intersection of the two straight-line portions of the conductance versus volume of titrant graph.



Graph

Reaction:

$$HCl+NaOH \rightarrow NaCl+H2O$$

At the equivalence point:

$N_1V_1=N_2V_2$

Where N_1, V_1 are the normality and volume of the acid, and N_2,V_2 are those of the base.

Conductometric titration is based on the principle that the electrical conductance of a solution depends on the concentration and mobility of ions present in it. During a titration, the ions in the solution are progressively replaced by other ions having different ionic mobilities. Therefore, the conductance changes as the titrant is added.

For the titration of **HCl with NaOH**:

- Initially, the solution contains highly mobile **H^+ ions**, so its conductance is high.
- As NaOH is added, **H^+ ions are neutralized** and replaced by the less mobile **Na^+ ions**.
- Hence, the conductance **decreases** up to the equivalence point.
- After the equivalence point, excess NaOH introduces **OH^- ions**, which have high ionic mobility.
- Therefore, the conductance **increases**.

The **equivalence point** is obtained from the intersection of the two straight-line portions

of the graph plotted between **conductance and volume of titrant added**.

Procedure :

- **Preparation of the solution:**

  Take a known volume, usually **20 ml of the given HCl solution**, in a clean beaker using a pipette.

- **Addition of distilled water:**

  Add a 20 ml of distilled water to the solution so that the conductivity cell is completely immersed. The addition of water does not affect the equivalence point because it does not change the number of moles of HCl present.

- **Setting up the conductivity meter:**

  Connect the conductivity cell to the conductivity meter. Wash the conductivity cell thoroughly with distilled water and rinse it with a small portion of the HCl solution.

- **Initial conductance:**

  Immerse the conductivity cell in the HCl solution and record the **initial conductance**.

- **Titration with NaOH:**

  Fill the burette with the standard **NaOH solution**. Add NaOH to the HCl solution in small portions, usually **0.5 mL at a time**.

- **Mixing:**

  After each addition of NaOH, stir the solution with a glass rod gently and allow the conductance reading to become stable.

- **Recording readings:**

Record the conductance after every addition of NaOH. Near the expected equivalence point, add NaOH in smaller portions, such as **0.5 mL**, to obtain accurate results.

- **Continue the titration:**

  Continue adding NaOH beyond the equivalence point and record sufficient readings.

- **Plot the graph:**

  Plot a graph with:

  - **X-axis:** Volume of NaOH added (mL)
  - **Y-axis:** Conductance (or corrected conductance)

- **Determination of equivalence point:**

  The conductance decreases initially and reaches a minimum at the equivalence point. After the equivalence point, the conductance increases due to the presence of excess OH^- ions.

- **Calculation:**

  The volume of NaOH corresponding to the intersection of the two straight-line portions of the graph is taken as the **equivalence-point volume**.

Tabular Form:

S.No	Volume of NaoH added (in ml)	Observed Conductance (m mho)
1.	0.0	
2	0.5	
3.	1.0	
4.	1.5	
5.	2.0	
6.	2.5	
7.	3.0	
8.	3.5	
9.	4.0	
10.	4.5	
11.	5.0	
12.	5.5	

13.	6.0	
14.	6.5	
15.	7.0	
16.	7.5	
17.	8.0	
18.	8.5	
19.	9.0	
20.	9.5	
21.	10.0	

Calculation:

hence HCl and NaOH react in a **1:1 ratio**, at the equivalence point:

$N_1V_1=N_2V_2$

Where:

- N_1 = Normality of HCl (To be determined
- V_1= Volume of HCl taken (20.0 ml)
- N_2= Normality of NaOH (1.0 N)
- V_2 = Volume of NaOH required at equivalence point
- (from the intersection point of the graph) = ---------- ml

Therefore,

$N_1 = N_2V_2 / V_1$

Normality of HCl obtained = --------- N

If the **strength of HCl in g/L** is required:

Strength=N × Equivalent weight

For HCl, equivalent weight = **36.5 g/eq**

Hence,

Strength of HCl = ---------- N HCl×36.5g/L

REPORT:

- **Normality of Hcl = ---------- N**
- **Strength of Hcl = ---------- g/L**

VIVA QUESTIONS

Q1. What is conductometric titration?

Ans. Conductometric titration is a titration in which the conductance of the solution is measured as the titrant is added to determine the equivalence point.

Q2. **What is the principle of conductometric titration?**

Ans**.** It is based on the change in electrical conductance due to changes in the concentration and mobility of ions during the reaction.

Q3. **Why is conductance measured in this titration?**

Ans. Because the concentration and mobility of ions change during neutralization, causing a measurable change in conductance.

Q4. **Which acid and base are used in this experiment?**

Ans. HCl (strong acid) and NaOH (strong base).

Q5.**What is the reaction between HCl and NaOH?**

Ans. $HCl+NaOH \rightarrow NaCl+H2O$

Q6.**Which ions are responsible for the high initial conductance?**

Ans. Mainly H^+ ions, because they have exceptionally high ionic mobility.

Q7. **Why does conductance decrease initially?**

Ans. H^+ ions are neutralized by OH^- ions and are replaced by the less mobile Na^+ ions.

Q8. **Why does conductance increase after the equivalence point?**

Ans. Excess NaOH introduces OH^- ions, which have high ionic mobility, causing conductance to increase.

Q9. **What is the equivalence point?**

Ans. It is the point at which the acid and base have reacted in their stoichiometric proportions.

Q10.**How is the equivalence point determined?**

Ans. By plotting conductance versus volume of NaOH added and finding the intersection of the two straight-line portions.

Q11. **Why is no indicator required?**

Ans. The equivalence point is determined from the conductance graph, so a chemical indicator is unnecessary.

Q12. **What is electrical conductance?**

Conductance is the reciprocal of resistance.

Q13.**What is the SI unit of conductance?**

Ans. Siemens (S) or milli mhos

Q14. **What is specific conductance?**

Ans. Specific conductance is the conductance of a solution contained between electrodes of 1 cm length and 1 cm^2 cross-sectional area.

Q15. **What is molar conductivity?**

Molar conductivity is the conductance of a solution containing **one mole of electrolyte**, placed between electrodes at unit distance.

Q16.**Why is distilled water added during the experiment?**

Ans. It helps to ensure that the conductivity cell is completely immersed. It does not change the number of moles of acid and hence does not change the equivalence-point volume.

Q17.**Why should the solution be stirred after every addition of NaOH?**

Ans. To ensure uniform mixing of the solution and obtain a stable conductance reading.

Q18.**Why should NaOH be added slowly near the equivalence point?**

Ans. To determine the equivalence point more accurately.

Q19.**What type of graph is obtained for strong acid–strong base titration?**

Ans. A V-shaped graph, with minimum conductance near the equivalence point.

Q20. **Why is HCl called a strong electrolyte?**

Ans. Because it undergoes almost complete ionization in aqueous solution.

Q21.**Why is NaOH called a strong electrolyte?**

Ans. Because it dissociates almost completely into Na^+ and OH^- ions in aqueous solution.

Q22. **What is the conductivity cell?**

Ans. It is an arrangement containing two electrodes used to measure the conductance of a solution.

Q23.**Which electrodes are commonly used in a conductivity cell?**

Ans. Platinized platinum electrodes are commonly used.

Q24. **Why are platinum electrodes used?**

Ans. Platinum is chemically inert and provides a stable conducting surface.

Q25. **What is cell constant?**

Ans. Cell constant=l/A, where l is the distance between electrodes and A is the electrode surface area.

Q26. **What is the relationship between conductance and resistance?**

Ans. G=1/R

Q27. **What happens to resistance when conductance increases?**

Ans. Resistance decreases, because conductance and resistance are reciprocals.

Q28. **Why is conductometric titration useful?**

Ans. It is useful for colored or turbid solutions where visual indicators may not give a clear endpoint.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 5

POTENTIOMETRIC TITRATION OF FERROUS FE(II) ION WITH POTASSIUM DICHROMATE ($K_2CR_2O_7$)

Aim:

To determine the **strength/concentration of a given Fe(II) solution** by carrying out a **potentiometric redox titration** using a standard Potassium Dichromate ($K_2Cr_2O_7$) solution, by measuring the change in **electrode potential (EMF)** during the titration and locating the **equivalence point** from the potentiometric titration curve.

Apparatus:

1. 50.0 ml Burette -1
2. 10.0 ml Pippette -1
3. 100 ml Beaker -1
4. 250 ml Beaker – 1
5. Platinum electrode – 1
6. Saturated calomel electrode – 1
7. Glass Stirror-1
8. Potetentiometer – 1

Solutions Required:

1. 0.1 N Potassium Dichroimate Solution ($K_2Cr_2O_7$) – 100 ml
2. 0.1 N Ferrous Fe(II) Ion solution – 100 ml
3. Acid mixture solution-5 ml
4. Distilled water

Principle:

Potentiometric titration is based on measuring the change in electrode potential during a redox reaction without using an indicator. In the titration of **potassium dichromate ($K_2Cr_2O_7$) against ferrous iron (Fe^{2+}) solution**, the potential changes sharply near the equivalence point.

The reaction is carried out in an **acidic medium**, usually using dilute sulfuric acid.

Redox reaction

Potassium dichromate acts as the **oxidizing agent**, while ferrous ions act as the **reducing**

agent:

$Cr_2O_7 + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$

Thus, **1 mole of $K_2Cr_2O_7$ reacts with 6 moles of Fe^{2+}**.

Here:

- **$Fe^{2+} \rightarrow Fe^{3+} + e^-$** — oxidation of ferrous ions
- **$Cr_2O_7^{2-} \rightarrow Cr^{3+}$** — reduction of dichromate ions

Potentiometric principle

A **platinum electrode** is immersed in the titration solution along with a suitable reference electrode. The platinum electrode responds to the oxidation–reduction potential of the solution.

As potassium dichromate is added to the ferrous iron solution:

1. Initially, the solution contains predominantly **Fe^{2+}/Fe^{3+}** redox couple.
2. During titration, Fe^{2+} is progressively oxidized to Fe^{3+}.
3. Near the **equivalence point**, a small addition of dichromate causes a **large change in electrode potential**.
4. The equivalence point is determined from the **sudden change in potential (ΔE)** or from the maximum value of **ΔE/ΔV**.
5. No visual indicator is required.

In one sentence

The principle of potentiometric titration of potassium dichromate and iron solution is based on measuring the change in redox potential of the Fe^{2+}/Fe^{3+} and $Cr_2O_7^{2-}/Cr^{3+}$ systems during the reaction, with the equivalence point being indicated by a sharp change in potential.

Experimental Procedure

1. **Fill the burette** with the standard potassium dichromate ($K_2Cr_2O_7$) solution. Remove any air bubbles from the burette tip.

2. **Pipette out a known volume** (e.g., **10.0 mL**) of the ferrous iron solution into a clean beaker.
3. Add about **5 mL** to provide the acidic medium required for the redox reaction.
4. Immerse the **platinum electrode and reference electrode** into the solution. Ensure that the electrodes are properly connected to the potentiometer / millivoltmeter.
5. Stir the solution gently and record the **initial potential (E)** before adding potassium dichromate.
6. Add potassium dichromate solution from the burette in **0.5 mL portions**. After each addition, stir the solution thoroughly and allow the potential to stabilize.
7. Record the corresponding **potential (E)** after each addition of $K_2Cr_2O_7$.
8. As the equivalence point is approached, add the dichromate solution in **smaller portions, such as 0.5 ml** because the potential changes rapidly near the endpoint.
9. Continue the titration for a few readings beyond the equivalence point.
10. Calculate and identify the volume corresponding to the **maximum change in potential**. This volume represents the **equivalence point**.

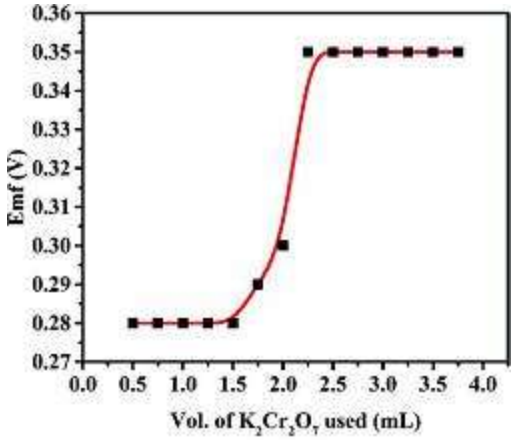
Observation Table

Endpoint: The volume of $K_2Cr_2O_7$ corresponding to the **maximum potential jump observed** is taken as the equivalence-point volume.

S.No	Volume of $K_2Cr_2O_7$ added (in ml)	Potential Observed(in Volts	S.No	Volume of $K_2Cr_2O_7$ added (in ml)	Potential Observed(in Volts
1.	0.0		16.	7.5	
2.	0.5		17.	8.0	
3.	1.0		18.	8.5	
4.	1.5		19.	9.0	
5.	2.0		20.	9.5	
6.	2.5		21.	10.0	
7.	3.0		22.	10.5	
8.	3.5		23.	11.0	
9.	4.0		24.	11.5	
10.	4.5		25.	12.0	

11.	5.0		26.	12.5	
12.	5.5		27.	13.0	
13.	6.0		28.	13.5	
14.	6.5		29.	14.0	
15.	7.0		30.	14.5	

Graph :



Calculations:

For a **potentiometric titration**, the concentration of the unknown solution is calculated from the **equivalence point volume** obtained from the titration curve.

Basic formula

At the equivalence point:

$N_1V_1=N_2V_2$

Where:

- N_1 = Normality of standard $K_2Cr_2O_7$ solution = 0.1N
- V_1 = Volume of standard solution used = 10.0 ml
- N_2 = Normality of unknown Fe(II) solution (To be determined)
- V_2 = Volume of unknown solution taken (Observed from Jump in the graph)

Therefore,

$N_2=N_1V_1 / V_2$

Report :

Normality of Unknown Fe (II) solution = N

Strength of Unknown Fe (II) in the given solution = Normality × Equivalent weight g/eq

Since, the equivalent weight of Fe^{2+} = 55.85/1 = 55.85 g/equiv.

Strength of Unknown Fe (II) in the given solution = N × 55.85 g/eq

= g/eq.

VIVA QUESTIONS

1. What is the aim of the experiment?

Answer: To determine the concentration/strength of the given **Fe(II) solution** by potentiometric titration using standard **potassium dichromate ($K_2Cr_2O_7$)** solution.

2. What type of titration is this?

Answer: It is a **potentiometric redox titration**.

3. What is the principle of the experiment?

Answer: **Fe^{2+} is oxidized to Fe^{3+}, while dichromate ions are reduced to Cr^{3+} in an acidic medium. The potential of the solution changes sharply near the equivalence point, which is used to determine the end point.**

4. What is the oxidizing agent?

Answer: Potassium dichromate, $K_2Cr_2O_7$**, is the oxidizing agent.**

5. What is the reducing agent?

Answer: Fe^{2+} (ferrous ion) **is the reducing agent.**

6. What happens to Fe^{2+} during the reaction?

Answer: Fe^{2+} loses one electron and is oxidized to Fe^{3+}.

$$Fe^{2+} \rightarrow Fe^{3+} + e^-$$

7. What happens to dichromate?

Answer: Dichromate ion gains electrons and is reduced to Cr^{3+}.

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

8. What is the overall ionic equation?

Answer: $Cr_2O_7 + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$

9. What is the stoichiometric ratio between Fe^{2+} and dichromate?

Answer: 1 mole of $K_2Cr_2O_7$ reacts with 6 moles of Fe^{2+}.

Therefore, $Fe2^+$:$K_2Cr_2O_7$=6:1

10. Why is acidic medium required?

Answer: H^+ ions are required for the reduction of dichromate to Cr^{3+}. Therefore, the reaction must be carried out in a sufficiently acidic medium.

11. Which acid is generally used?

Answer: a dilute **sulfuric acid (H_2SO_4)** is generally used.

12. Why is HCl not preferred?

Answer: Chloride ions can undergo oxidation under strongly oxidizing conditions, causing interference and inaccurate results.

13. Why is nitric acid not preferred?

Answer: Nitric acid itself is an oxidizing acid and can oxidize Fe^{2+}, producing an error in the determination.

14. Why is potassium dichromate considered a primary standard?

Answer: It is available in high purity, is stable, has a relatively high molar mass, is not highly hygroscopic, and can be weighed accurately to prepare a standard solution.

15. What is the equivalent weight of $K_2Cr_2O_7$ in this reaction?

Answer: Molar mass of $K_2Cr_2O_7$ ≈ **294.18 g mol^{-1}**

Number of electrons involved = 6.

Equivalent weight=294.18/6

16. What is the equivalent weight of Fe^{2+}?

Answer: Fe^{2+} changes to Fe^{3+} by losing one electron.

Equivalent weight=55.85/1

17. Why is potentiometry used instead of an ordinary indicator?

Answer: In potentiometric titration, the end point is determined from the **change in electrode potential**, so no visual indicator is required.

18. What is measured during the experiment?

Answer: The **potential difference (EMF)** between the indicator electrode and reference electrode is measured after successive additions of $K_2Cr_2O_7$.

19. Which electrode is commonly used as the indicator electrode?

Answer: A **platinum electrode** is commonly used.

20. Why is platinum used?

Answer: Platinum is chemically inert and provides a surface for the oxidation-reduction reactions without participating in the reaction.

21. What is the role of the reference electrode?

Answer: It provides a stable and known reference potential against which the potential of the indicator electrode is measured.

22. What is the end point in potentiometric titration?

Answer: It is the point corresponding to the **equivalence point**, where stoichiometrically equivalent amounts of Fe^{2+} and dichromate have reacted.

23. How is the end point determined from the readings?

Answer: Plot **potential (E) versus volume of $K_2Cr_2O_7$ added**. The end point is obtained from the **sharp change in potential**, or more accurately from the maximum value of:

24. Why does the potential change sharply near the end point?

Answer: Near the equivalence point, the concentrations of the oxidized and reduced species change rapidly, causing a large change in the electrode potential.

25. What is the Nernst equation?

Answer:

$E=E^{\circ}-0.0591/n\log\ [Red][Ox]$ at 25°C.

26. Why should the solution be stirred?

Answer: Stirring ensures uniform mixing and allows the electrode potential to reach a stable value.

27. Why should the platinum electrode be cleaned?

Answer: To remove impurities or deposits that may affect electron transfer and cause incorrect potential readings.

28. What happens if Fe^{2+} gets oxidized by air before titration?

Answer: Some Fe^{2+} changes to Fe^{3+}, reducing the amount of Fe^{2+} available for reaction with dichromate. This can cause an **error in the calculated concentration**.

29. What is the oxidation state of chromium in $K_2Cr_2O_7$?

Answer: +6.

30. What is the oxidation state of chromium after the reaction?

Answer: It becomes **+3** in Cr^{3+}.

31. How many electrons are accepted by one dichromate ion?

Answer: Six electrons.

32. How many electrons are released by six Fe^{2+} ions?

Answer: Six Fe^{2+} ions each release one electron, so a total of **six electrons** are released.

33. Is $K_2Cr_2O_7$ a self-indicator?

Answer: No. Unlike $KMnO_4$, potassium dichromate does **not** provide a sufficiently intense self-indicating colour change at the equivalence point. In this experiment, the end point is determined potentiometrically.

34. What colour does dichromate have?

Answer: Dichromate ion is **orange**.

35. What colour is Cr^{3+} in solution?

Answer: Cr^{3+} solutions are generally **green or violet**, depending on the chemical environment.

36. What is the main advantage of potentiometric titration?

Answer: It gives an objective end point and is particularly useful for **coloured or turbid solutions**, where visual indicators may be difficult to use.

Students should especially remember these:

1. **Oxidizing agent:** $K_2Cr_2O_7$
2. **Reducing agent:** Fe^{2+}
3. **$Fe^{2+} \rightarrow Fe^{3+}$:** Oxidation
4. **$Cr^{6+} \rightarrow Cr^{3+}$:** Reduction

5. **Acid used:** H_2SO_4
6. **Indicator electrode:** Platinum
7. **Fe^{2+} : $Cr_2O_7^{2-}$ ratio: 6 : 1**
8. **Equivalent weight of Fe^{2+}: 55.85 g/equiv**
9. **Equivalent weight of $K_2Cr_2O_7$: 49.03 g/equiv**
10. **End point:** Sharp change in potential
11. **Graph:** Potential (E) vs volume of $K_2Cr_2O_7$
12. **Overall reaction: $Cr_2O_7+14H^++6Fe^{2+}\rightarrow 2Cr^{3+}+7H_2O+6Fe^{3+}$**

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 6

PREPARATION A BAKELITE

Aim: To prepare phenol-formaldehyde resin.

Required Apparatus: Beakers, conical flask, glass rod, measuring cylinders, fractionation weight box etc.

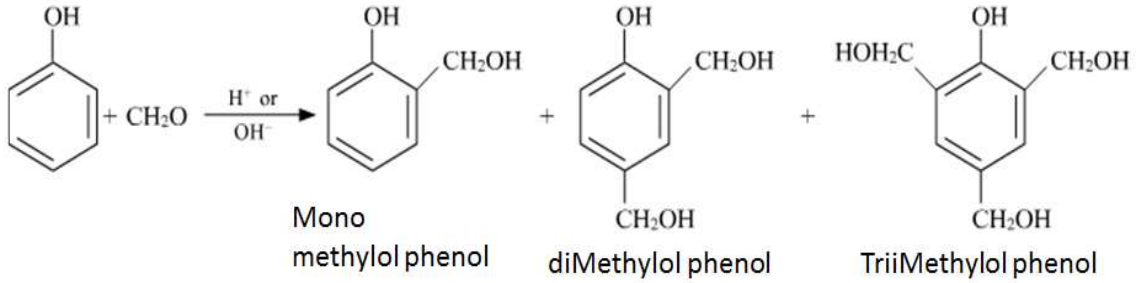
Chemicals: glacial acetic acid,40% formaldehyde solution, phenol, conc. HCl, distilled water.

Theory: phenol resins are condensation polymerization product of phenolic derivative with aldehyde (like formaldehyde). It is prepared by condensing phenol with formaldehyde in the presence of acid or alkaline catalyst.

Reaction:

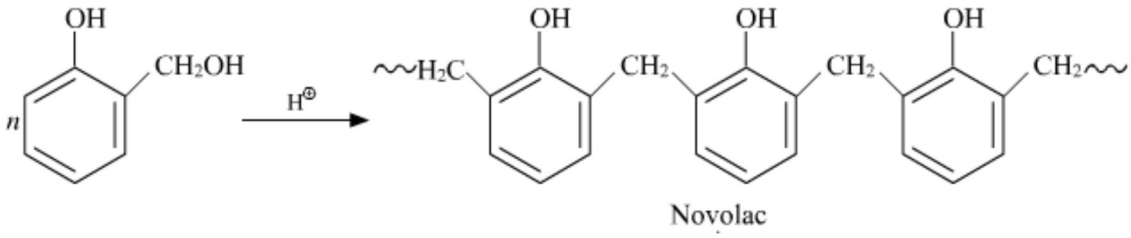
Step-1

The first step is reaction between phenol and formaldehyde, forms mono, di and tri-methylol phenols.



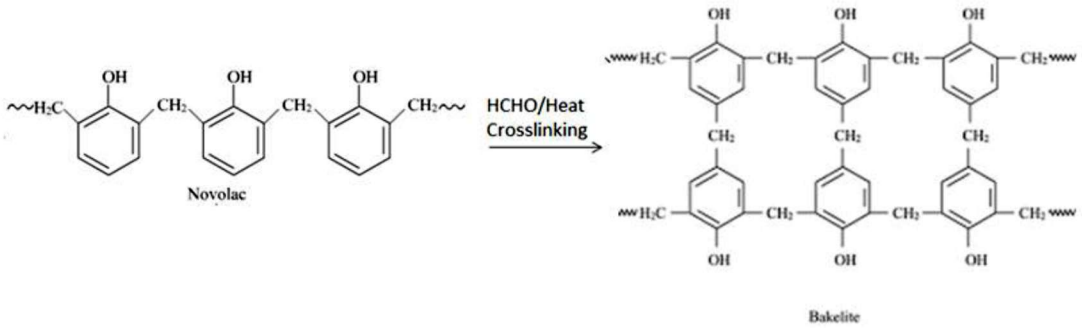
Step-II

When mehtylol phenols are heated with excess of phenol in presence of acid catalyst, the methylol phenols condense with phenol through methylene linkages to from novolac.



Step-III

Further heating methylol phenol and phenol (or) Resole and Phenol both in the presence of a curing agent (hexamethylene tetramine) products hard, regid, infusible, cross linked polymer is called bakelite.



Further heating of novolac in the presence of hexamethylene tetramine produces hard, rigid, infusible polymer called Bakelite.

Procedure:

1. Place 5ml of glacial acetic acid and 2.5ml of formaldehyde solution in 500ml beaker.
2. Add 2gm of phenol and 1ml of conc.HCl solution in it.
3. Heat the solution slowly with constant stirring for 5mins.
4. A large mass of pink plastic is formed.
5. The residue obtained is washed several times with distilled water.
6. Dry the product and calculate the yield accurately.

Properties:

1. It is insoluble in almost all organic solvents
2. It is water –resistant.
3. It has good electrical insulating character.
4. It is thermally stable.
5. It is highly resistant to non-oxidizing acids, salts and many organic s
6. It is attacked by alkalis because of presence of –OH groups.

Uses:

1. For making telephone parts, radio and T.V. cabinets
2. For making electrical equipment like switches, plugs, holders, switch boards, heaters handles, etc.
3. It is used in the preparation of paints and varnishes and protective coatings
4. In the production of ion–exchange resins

Result: The weight of obtained Bakelite is **gms.**

VIVA QUESTIONS

1. What is Bakelite?

Bakelite is a thermosetting phenol-formaldehyde resin.

2. Which monomers are used to prepare Bakelite?

Phenol and formaldehyde are used as monomers.

3. **What type of polymerization occurs in Bakelite formation?**

Condensation polymerization occurs with elimination of water.

4. **What is the role of formaldehyde?**

Formaldehyde acts as a cross-linking agent during polymer formation.

5.**What is the role of phenol?**

Phenol provides the aromatic units that form the polymer backbone.

6.**Why is Bakelite called a thermosetting polymer?**

It forms a highly cross-linked structure and cannot be softened or remoulded by heating.

7. **What is the function of the catalyst?**

The catalyst promotes the condensation reaction between phenol and formaldehyde.

8. **What is novolac?**

Novolac is a linear or slightly branched phenol-formaldehyde resin formed under acidic conditions with excess phenol.

9. **Why does Bakelite have high thermal and electrical resistance?**

Its rigid, highly cross-linked structure provides good heat resistance and electrical insulation.

10. **Mention two applications of Bakelite.**

It is used for electrical switches, plugs, sockets, and handles of heat-resistant utensils.

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 7

PREPARATION OF NANOMATERIALS BY PRECIPITATION METHOD

Aim : To prepare the Magnetite Nanoparticles

Apparatus : 100 ml Glass Beaker, 5ml Measuring jar, 50ml Burette, Magnetic Stirrer, Magnet, Wash bottle, Glass rod, Funnel, Electronic balance etc.

Chemicals : l M $FeCl_3$, 2M $FeSO_4$, l M NH_3 solution

Procedure:

In a 100 ml Glass beaker, add 5 ml of lM $FeCl_3$ and 1 ml of 2M $FeSO_4$ solution, place a magnet under the container and stir continuously using magnetic stirrer. Add about 1 ml of 1 M NH_3 up to 5 ml every 10 seconds over a period of 5-10 minutes (**Do not add the solution all at once**). Then Black particles begin to form after the continuous stirring for 30 more minutes. Let the solution settle down after stirring. Remove the clear solution without losing the solid with the help of a filter paper and funnel.

$$FeCl_3 + 3\ NH_4OH \rightarrow Fe(OH)_3 + 3\ NH_4Cl$$

$$FeSO_4 + 2\ NH_4OH \rightarrow Fe(OH)_2 + (NH_4)_2\ SO_4$$

Oxidation at pH 9-10

$$Fe(OH)_x^{3-x} + Fe(OH)_Y^{2-Y} \longrightarrow Fe_3O_4 \text{ (Magnetite)}$$

Wash the precipitate for three times with the distilled water. Dry the precipitate in an oven for 2 hours. Examine the spikes that forms as you move the magnet around. Finally collect the obtained **spikes of Magnetite Nanoparticles** and calculate the weight of Magnetite Nanoparticle by using electronic balance.

Report : The weight of obtained Magnetite Nanoparticles =_____ gm

VIVA QUESTIONS

1. **What are iron nanomaterials?**
   Iron nanomaterials are iron-based particles having at least one dimension in the nanoscale range, typically 1–100 nm.
2. **What is the precipitation method?**
   It is a chemical method in which dissolved iron ions are converted into an insoluble solid by adding a precipitating agent.
3. **Which iron salts are commonly used as precursors?**
   Ferric chloride ($FeCl_3$) and ferrous sulfate ($FeSO_4$) or ferrous chloride ($FeCl_2$) are commonly used.
4. **What is the role of the precipitating agent?**
   It reacts with iron ions to form an insoluble iron hydroxide or oxide precipitate.
5. **Why is pH important in the preparation?**
   pH controls the precipitation reaction and strongly affects the particle size, morphology, and phase of the nanomaterial.
6. **Why is stirring required during precipitation?**
   Continuous stirring ensures uniform mixing and helps produce a more homogeneous precipitate.
7. **Why is the precipitate washed?**
   Washing removes unwanted soluble ions, excess reactants, and impurities from the precipitate.
8. **Why is the precipitate dried and heated?**
   Drying removes moisture, while heating can convert the precursor hydroxide into the desired iron oxide nanomaterial.
9. **What factors affect the size of iron nanoparticles?**
   Concentration, pH, temperature, stirring rate, reaction time, and rate of addition of the precipitating agent affect particle size.
10. **Mention two applications of iron nanomaterials.**
    They are used in magnetic separation, environmental remediation, catalysis, sensors, and biomedical applications.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 8

DETERMINATION OF CORROSION RATE BY WEIGHT LOSS METHOD

AIM

To determine the corrosion rate of a given mild steel specimen in an acidic medium (1 M HCl) at room temperature using the weight loss method.

APPARATUS REQUIRED

- Mild steel coupons/specimens
- 1 M Hydrochloric acid (HCl) solution
- 250 mL beakers
- Digital analytical balance (precision ±0.1 mg)
- Vernier caliper
- Emery paper/Sandpaper of different grades
- Acetone or ethanol
- Glass rod
- Distilled water

PRINCIPLE

Corrosion is the gradual destruction or deterioration of a metal due to its chemical or electrochemical reaction with the surrounding environment.

When a mild steel specimen is immersed in an acidic solution such as 1 M HCl, an electrochemical corrosion reaction takes place.

Anodic reaction (Oxidation):

$Fe \rightarrow Fe^{2+} + 2e^-$

Cathodic reaction (Reduction):

$2H^+ + 2e^- \rightarrow H_2\uparrow$

During corrosion, iron atoms dissolve into the solution and the metal specimen loses weight. The corrosion rate can therefore be determined by measuring the loss in weight of the specimen, its exposed surface area, and the duration of exposure.

PROCEDURE

1. Take a clean mild steel coupon and polish its surface using emery paper to remove rust, oxide layers, and other impurities.
2. Wash the specimen thoroughly with distilled water.
3. Rinse the specimen with acetone or ethanol to remove oil, grease, and other contaminants.

4. Allow the specimen to dry completely.
5. Measure the length, width, and thickness of the specimen.
6. Calculate the total surface area of the specimen using the measured dimensions.
7. Weigh the clean and dry specimen accurately using a digital analytical balance. Record this value as the initial weight (W_1).
8. Take 100 mL of 1 M HCl solution in a clean 250 mL beaker.
9. Suspend the mild steel specimen completely in the HCl solution using a suitable thread or glass hook.
10. Keep the specimen immersed in the solution for a fixed period, such as 24 or 48 hours. Record the exact exposure time (t).
11. After the required exposure time, carefully remove the specimen from the solution.
12. Gently clean the specimen under running water to remove loose corrosion products.
13. Rinse the specimen with acetone or ethanol and dry it completely.
14. Weigh the dried specimen again using the analytical balance. Record this value as the final weight (W_2).
15. Calculate the weight loss and corrosion rate using the formulas given below.

OBSERVATION TABLE

Specimen No.	**Length, l (cm)**	**Width, w (cm)**	**Thickness, d (cm)**	**Surface Area, A (cm^2)**	**Initial Weight, W_1 (mg)**	**Final Weight, W_2 (mg)**	**Weight Loss, W (mg)**	**Time, t (hours)**
1								
2								

CALCULATIONS

A. Total Surface Area

For a rectangular mild steel specimen:

$$\mathbf{A = 2[(l \times w) + (w \times d) + (l \times d)]\ cm^2}$$

Where:

- **l** = Length of the specimen in cm
- **w** = Width of the specimen in cm
- **d** = Thickness of the specimen in cm
- **A** = Total surface area in cm^2

B. Weight Loss

The weight loss of the specimen is calculated as:

$$\mathbf{W = W_1 - W_2}$$

Where:

- $\mathbf{W_1}$ = Initial weight of the specimen in mg
- $\mathbf{W_2}$ = Final weight of the specimen in mg
- **W** = Weight loss in mg

C. Corrosion Rate

The corrosion rate is calculated using the following formula:

$$\mathbf{CR = (87.6 \times W) / (\rho \times A \times t)}$$

Where:

- **CR** = Corrosion rate in mm/year
- **W** = Weight loss in mg
- **ρ** = Density of the metal in g/cm^3
- **A** = Total exposed surface area in cm^2
- **t** = Exposure time in hours
- **87.6** = Conversion factor for obtaining the corrosion rate in mm/year

For mild steel:

$$\mathbf{\rho = 7.85\ g/cm^3}$$

7. RESULT

The weight loss of the mild steel specimen in 1 M HCl is found to be:

Weight loss = _____ mg

The corrosion rate of the mild steel specimen is found to be:

Corrosion rate = _____ mm/year

VIVA QUESTIONS AND ANSWERS

1. What is corrosion?

Corrosion is the gradual degradation or destruction of a metal due to its chemical or electrochemical reaction with the surrounding environment.

2. Why is acetone used before weighing the metal specimen?

Acetone is used as a degreasing agent. It removes oil, grease, and other contaminants from the metal surface, ensuring accurate weight measurement.

3. What type of corrosion occurs when mild steel is immersed in an acidic solution?

Electrochemical corrosion occurs. Small anodic and cathodic regions are formed on the metal surface, resulting in the dissolution of the metal.

4. How does an increase in temperature affect the corrosion rate?

An increase in temperature generally increases the corrosion rate because it increases the kinetic energy of the reacting particles and speeds up the corrosion reactions.

5. What is the purpose of the constant 87.6 in the corrosion rate formula?

The value 87.6 is a conversion factor used to convert the measured weight loss, surface area, density, and exposure time into the corrosion rate expressed in millimetres per year (mm/year).

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 9

MEASUREMENT OF PH AND ELECTRICAL CONDUCTANCE OF DEIONISED WATER

AIM

To determine the pH and electrical conductance of a given sample of deionised water at room temperature using a digital pH meter and a conductivity meter.

APPARATUS REQUIRED

- Digital pH meter equipped with a glass combination electrode
- Digital Conductivity meter equipped with a conductivity cell (platinum electrode)
- Clean glass beakers (100 mL, 250 mL)
- Wash bottle filled with distilled water (for rinsing probes)
- High-quality lint-free tissue wipes
- Standard reference buffer solutions (pH 4.0, pH 7.0, and pH 9.2) for pH calibration
- Standard potassium chloride (KCl) solution (0.01 M) for conductivity calibration
- Fresh sample of deionised water

PRINCIPLE

pH Measurement

The term pH stands for 'potential of Hydrogen' and represents the negative logarithm (to the base 10) of the hydrogen ion concentration in a solution:

$$pH = -\log_{10}[H^+]$$

Pure water undergoes self-ionization to a minor extent: $H_2O \rightleftharpoons H^+ + OH^-$. At 25°C, the concentration of hydrogen ions in highly pure water is 10^{-7} M, resulting in an exact neutral pH of 7.0. The digital pH meter records the potential difference established between the sensitive glass membrane electrode and a stable reference electrode inside the solution, directly processing this voltage into a human-readable pH value.

Electrical Conductance Measurement

Electrical Conductance (G) defines the ease with which electric current streams through a liquid sample. It functions as the reciprocal of electrical resistance ($G = 1/R$) and is quantified in Siemens (S) or Micro-Siemens (μS). Pure water exhibits an exceptionally low conductivity because it lacks dissolved chemical elements.

Deionised (DI) water is stripped of dissolved mineral salts (such as Na^+, Ca^{2+}, and Cl^-) using specialized ion-exchange columns. Consequently, high-purity deionised water shows negligible electrical conductance (typically 0.5 to 5.0 μS/cm). The conductivity meter operates by applying

an alternate voltage across two parallel plates in the submerged cell, evaluating the resulting current.

PROCEDURE

Part A: Calibration and Measurement of pH

1. Power on the digital pH meter and allow the electronic circuits to warm up for 15 minutes.
2. Carefully extract the glass combination electrode from its storage container, rinse it thoroughly with a jet of distilled water from a wash bottle, and gently blot dry with lint-free tissue paper (avoid rubbing to prevent generating static electricity).
3. Submerge the electrode into the standard pH 7.0 buffer solution. Allow the display numbers to settle, then adjust the calibration/buffer control knob until the screen reads precisely 7.00.
4. Remove the electrode, rinse it again with distilled water, blot dry, and test against either pH 4.0 or pH 9.2 buffer solutions to confirm accurate slope adjustments.
5. Rinse the validated electrode a final time. Pour roughly 50 mL of your deionised water sample into a pristine 100 mL beaker, submerge the probe, stir smoothly with a glass rod, let the value stabilize, and note the pH display.

Part B: Calibration and Measurement of Conductance

1. Turn on the digital conductivity meter and configure the target settings.
2. Rinse the platinum conductivity cell thoroughly using distilled water to eliminate residual ions, then gently shake off excess droplets.
3. Calibrate the conductivity meter unit by dipping the cell into a standard reference 0.01 M KCl solution if dictated by laboratory protocol, ensuring the internal temperature coefficient settings match ambient conditions.
4. Cleanse the calibrated conductivity cell thoroughly with distilled water to prevent cross-contamination.
5. Pour a fresh 50 mL portion of the deionised water sample into a separate clean beaker. Immerse the conductivity cell entirely, verify that no air pockets remain trapped within the active slot of the probe, wait for the display to settle, and record the electrical conductance in µS/cm.

TABULAR FORM

Record the experimental findings in the structured table below:

Parameter Measured	Trial 1	Trial 2	Trial 3	Average Value
pH of Deionised Water				
Electrical Conductance (µS/cm)				

CALCULATION & INTERPRETATION

1. pH Readings: No complex calculations are necessary. Take the mathematical average of the stable numbers gathered across the three individual trials.

2. Conductance Adjustments: If the equipment outputs raw electrical conductance (G), find the true specific conductivity (K) by applying the formula:
Specific Conductivity (K) = Measured Conductance (G) × Cell Constant (θ)

Note on Water Quality: Pure deionised water lacks trace minerals to block incoming gasses. Upon brief air exposure, it actively dissolves atmospheric carbon dioxide (CO_2) to yield weak carbonic acid (H_2CO_3). Consequently, freshly collected DI water typically features a minor down-shift in pH, resting between 5.5 and 6.8.

RESULTS

The pH of the tested deionised water sample is found to be: _____
The electrical conductance of the tested deionised water sample is: _____ µS/cm

---

VIVA QUESTIONS

1. What differentiates distilled water from deionised water?

Distilled water is generated through thermal phase change (boiling into gas and cooling back to liquid), which separates volatile compounds and solid minerals. Deionised water is filtered through cation and anion exchange resins which target and extract dissolved mineral ions specifically. DI water may still retain microscopic non-ionic particles or organic bacteria if untreated by secondary carbon filters.

2. Why does pure deionised water show an acidic shift when exposed to normal atmospheric air?

Because deionised water is stripped of mineral carbonates, it has zero buffering capacity. It quickly absorbs carbon dioxide (CO2) from ambient air, triggering a chemical reaction that yields a dilute solution of weak carbonic acid (H2CO3), thereby dropping the pH below 7.0.

3. Why must we avoid scrubbing or rubbing the fragile bulb of a pH electrode with tissue wipes?

Rubbing the thin glass outer membrane generates a localized static electrical charge. This artificial static voltage distorts the minute millivolt signal read by the instrument assembly, causing highly unstable, fluctuating, or inaccurate readings.

4. What are the standard engineering units used to express electrical conductivity in water analysis?

The primary international SI unit is Siemens per metre (S/m). However, because pure water and environmental water samples possess exceptionally low baseline conductivity values, industrial

and laboratory professionals record findings in Micro-Siemens per centimetre (µS/cm).

5. Explain why deionised water serves as an exceptional electrical insulator.

Electrical current requires physical ionic charge carriers (cations and anions) to migrate through a liquid medium. Because the deionisation sequence filters out virtually all dissolved mineral elements, there are almost no free ions remaining to convey the electrical charge, rendering its conductance close to zero.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT - 10

DETERMINATION OF EMF OF DANIELL CELL AND VERIFICATION OF NERNST EQUATION

AIM

To determine the electromotive force (EMF) of a Daniell cell and to verify the Nernst equation by studying the variation of EMF with a change in concentration of the electrolytes.

APPARATUS REQUIRED

- Two clean beakers (100 mL or 250 mL)
- Zinc strip (electrode) and Copper strip (electrode)
- Salt bridge (or U-tube filled with agar-agar and KCl / KNO_3)
- digital multimeter / high-resistance voltmeter
- Connecting wires with crocodile clips
- Sandpaper
- Measuring cylinder / pipette
- Stand and clamps (if required)

CHEMICALS REQUIRED

- Zinc sulphate ($ZnSO_4$) solutions of different molarities: 1.0 M, 0.5 M, 0.1 M, 0.05 M, 0.01 M, etc.
- Copper sulphate ($CuSO_4$) solutions of different molarities: 1.0 M, 0.5 M, 0.1 M, 0.05 M, 0.01 M, etc.
- Agar-agar (for preparing salt bridge)
- Potassium chloride (KCl) or Potassium nitrate (KNO_3) saturated solution
- Distilled water

PRINCIPLE

A Daniell cell is a galvanic (voltaic) cell in which zinc acts as the anode and copper acts as the cathode. The cell representation is:

$$\mathbf{Zn\ (s)\ |\ Zn^{2+}\ (aq)\ ||\ Cu^{2+}\ (aq)\ |\ Cu\ (s)}$$

The cell reaction is:

$$\mathbf{Zn\ (s) + Cu^{2+}\ (aq) \rightarrow Zn^{2+}\ (aq) + Cu\ (s)}$$

The standard EMF of the Daniell cell (E°cell) is 1.10 V at 25 °C, calculated as:

$$\mathbf{E°cell = E°(Cu^{2+}/Cu) - E°(Zn^{2+}/Zn) = 0.34 - (-0.76) = 1.10\ V}$$

According to the Nernst equation, the EMF of the cell under non-standard conditions is given by:

$$\mathbf{Ecell = E°cell - (RT / nF)\ ln\ ([Zn^{2+}] / [Cu^{2+}])}$$

At 25 °C (298 K), substituting the values of R, T and F, the equation simplifies to:

$$\mathbf{Ecell = E°cell - (0.0591 / n)\ log\ ([Zn^{2+}] / [Cu^{2+}])}$$

Since n = 2 for the Daniell cell reaction,

$$\mathbf{Ecell = 1.10 - (0.0591 / 2)\ log\ ([Zn^{2+}] / [Cu^{2+}]) = 1.10 - 0.02955\ log\ ([Zn^{2+}] / [Cu^{2+}])}$$

By measuring the EMF of the cell at different concentrations of $ZnSO_4$ and $CuSO_4$ and comparing the experimental values with those calculated from the Nernst equation, the validity of the Nernst equation can be verified.

PROCEDURE

1. Clean the zinc and copper strips thoroughly with sandpaper to remove any oxide layer and wash them with distilled water.
2. Prepare the salt bridge by filling a U-tube with a hot solution of agar-agar containing saturated KCl (or KNO_3) and allow it to solidify.
3. Take two clean beakers. Pour 1.0 M $ZnSO_4$ solution into one beaker and 1.0 M $CuSO_4$ solution into the other beaker.
4. Dip the zinc strip into the $ZnSO_4$ solution (anode) and the copper strip into the $CuSO_4$ solution (cathode).
5. Connect the two half-cells using the salt bridge. Ensure that the ends of the salt bridge dip into the respective solutions.
6. Connect the zinc electrode to the negative terminal and the copper electrode to the positive terminal of a high-resistance voltmeter (or digital multimeter).
7. Note the steady EMF reading of the cell. This corresponds to the standard concentration (1 M) of both electrolytes.
8. Replace the solutions with different combinations of concentrations of $ZnSO_4$ and $CuSO_4$ (keeping one concentration fixed while varying the other, or varying both).
9. For each combination, record the experimental EMF carefully after the reading stabilizes.
10. Calculate the theoretical EMF for each concentration using the Nernst equation and compare it with the experimental value.
11. Repeat the measurements to ensure reproducibility and calculate the mean values.

OBSERVATIONS / TABULAR FORM

Temperature of the laboratory = _____ °C

Table 1: Variation of EMF of Daniell Cell with Concentration

S. No.	$[ZnSO_4]$ (M)	$[CuSO_4]$ (M)	log ($[Zn^{2+}]/[Cu^{2+}]$)	Experimental EMF (V)	Calculated EMF (V)
1	1.0	1.0	0		1.10
2	1.0	0.5	0.3010		1.091
3	1.0	0.1	1.000		1.070
4	1.0	0.01	2.000		1.041
5	0.5	1.0	−0.3010		1.109
6	0.1	1.0	−1.000		1.130
7	0.01	1.0	−2.000		1.159
8	0.1	0.1	0		1.10

Note: Calculated EMF values are obtained using Ecell = 1.10 − 0.02955 log ($[Zn^{2+}]/[Cu^{2+}]$). Experimental columns are to be filled during the practical.

CALCULATION

The theoretical EMF for each concentration ratio is calculated using the Nernst equation at 25 °C:

$$\mathbf{E_{cell} = 1.10 - (0.0591 / 2) \log ([Zn^{2+}] / [Cu^{2+}])}$$

Example calculation:

For $[ZnSO_4]$ = 1.0 M and $[CuSO_4]$ = 0.1 M,

log ($[Zn^{2+}]/[Cu^{2+}]$) = log (1.0 / 0.1) = log 10 = 1

E_{cell} = 1.10 − 0.02955 × 1 = 1.10 − 0.02955 = 1.07045 V ≈ 1.070 V

Percentage error (if required) = |(Experimental EMF − Calculated EMF) / Calculated EMF| × 100

RESULTS

1. The EMF of the Daniell cell under standard conditions (1 M $ZnSO_4$ and 1 M $CuSO_4$) was found to be approximately _____ V (theoretical value = 1.10 V).

2. The experimental EMF values obtained at different concentrations of the electrolytes are in close agreement with the values calculated using the Nernst equation.

3. When the concentration of Cu^{2+} ions is decreased (keeping $[Zn^{2+}]$ constant), the EMF of the cell decreases.

4. When the concentration of Zn^{2+} ions is decreased (keeping $[Cu^{2+}]$ constant), the EMF of the cell increases.

5. The Nernst equation is verified, as the observed variation of cell EMF with concentration matches the theoretical predictions.

PRECAUTIONS

- Electrodes must be cleaned properly with sandpaper before use.
- Use a high-resistance voltmeter so that negligible current is drawn from the cell.
- The salt bridge should not contain air bubbles and both ends must dip properly into the solutions.
- Solutions of accurate concentrations should be prepared using distilled water.
- Readings should be taken only after the voltmeter shows a steady value.
- Avoid contamination of solutions; rinse electrodes with distilled water between measurements.

VIVA QUESTIONS

1. What is a Daniell cell?

Daniell cell is a primary galvanic cell in which zinc acts as anode and copper as cathode. The cell reaction is $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$. Its standard EMF is 1.10 V.

2. Write the cell representation of a Daniell cell.

Zn (s) | Zn^{2+} (aq) || Cu^{2+} (aq) | Cu (s)

3. What is the Nernst equation?

The Nernst equation relates the electrode potential (or cell EMF) to the standard electrode potential and the concentrations (or activities) of the species involved: $E = E° − (RT/nF) \ln Q$, where Q is the reaction quotient.

4. Write the Nernst equation for the Daniell cell at 25 °C.

$Ecell = 1.10 − (0.0591/2) \log ([Zn^{2+}]/[Cu^{2+}])$

5. What is the value of n in the Nernst equation for Daniell cell?

n = 2, because two electrons are transferred in the cell reaction.

6. Why is a salt bridge used in the Daniell cell?

The salt bridge maintains electrical neutrality in both half-cells by allowing the flow of ions and completes the circuit. It also prevents the mixing of the two electrolyte solutions.

7. What happens to the EMF of the Daniell cell if the concentration of Cu^{2+} is decreased?

According to the Nernst equation, decreasing $[Cu^{2+}]$ increases the value of $\log ([Zn^{2+}]/[Cu^{2+}])$, which decreases the EMF of the cell.

8. What happens to the EMF if the concentration of Zn^{2+} is decreased?

Decreasing $[Zn^{2+}]$ decreases $\log ([Zn^{2+}]/[Cu^{2+}])$, which increases the EMF of the cell.

9. Why is a high-resistance voltmeter preferred for measuring cell EMF?

A high-resistance voltmeter draws negligible current from the cell, so the measured potential is close to the true EMF (open-circuit potential).

10. What is the standard reduction potential of Zn^{2+}/Zn and Cu^{2+}/Cu electrodes?

$E°(Zn^{2+}/Zn) = -0.76$ V and $E°(Cu^{2+}/Cu) = +0.34$ V.

11. Is Daniell cell a primary or secondary cell?

Daniell cell is a primary cell (non-rechargeable).

12. What is the role of the porous pot or salt bridge in Daniell cell?

It allows the migration of ions to maintain charge balance without mixing of the two solutions.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 11

DETERMINATION OF CELL CONSTANT OF A CONDUCTIVITY CELL

AIM

To determine the cell constant of the given conductivity cell using a standard solution of potassium chloride (KCl) of known specific conductance.

APPARATUS REQUIRED

- Conductivity meter
- Conductivity cell (with platinum electrodes)
- Beakers (100 mL / 250 mL)
- Volumetric flasks (100 mL, 250 mL, 1000 mL)
- Pipette and measuring cylinder
- Thermometer
- Wash bottle
- Tissue paper / filter paper

CHEMICALS REQUIRED

- Potassium chloride (KCl) – analytical grade, dried
- Conductivity water (double distilled / de-ionised water of very low conductivity)
- Standard KCl solutions: 0.1 M, 0.01 M, 0.001 M (or 0.1 N, 0.01 N, etc.)

PRINCIPLE

The resistance (R) of an electrolyte solution between two electrodes is given by:

$$\mathbf{R = \rho \times (l / a)}$$

where ρ is the specific resistance, l is the distance between the electrodes and a is the area of cross-section of the electrodes.

The reciprocal of resistance is conductance (G):

$$\mathbf{G = 1 / R}$$

Specific conductance (κ) is the reciprocal of specific resistance:

$\kappa = 1 / \rho = (1 / R) \times (l / a) = G \times (l / a)$

The quantity (l / a) is called the cell constant (K or G*) of the conductivity cell.

Cell Constant (K) = $l / a = \kappa / G$

or

Cell Constant = Specific Conductance / Observed Conductance

The cell constant has the unit cm^{-1} (or m^{-1}). It cannot be measured accurately from the physical dimensions of the electrodes because of edge effects and imperfect geometry. Therefore, it is determined experimentally by measuring the conductance of a standard solution (usually KCl) whose specific conductance is accurately known at a given temperature.

Once the cell constant is known, the specific conductance of any unknown solution can be calculated as:

κ = Cell Constant × Observed Conductance

PROCEDURE

1. Switch on the conductivity meter and allow it to warm up for 10–15 minutes.
2. Prepare standard solutions of KCl (0.1 M, 0.01 M and 0.001 M) using conductivity water in volumetric flasks. (0.1 M KCl = 7.455 g KCl dissolved in conductivity water and made up to 1 litre.)
3. Note the room temperature (or the temperature of the solutions) accurately with a thermometer.
4. Rinse the conductivity cell thoroughly with conductivity water and then with the solution whose conductance is to be measured.
5. Take about 40–50 mL of 0.1 M KCl solution in a clean dry beaker. Dip the conductivity cell into the solution so that the electrodes are completely immersed. Ensure that no air bubbles are trapped between the electrodes.
6. Connect the cell to the conductivity meter and measure the conductance (or resistance). Record the stable reading.
7. Remove the cell, rinse it with conductivity water, and repeat the measurement with 0.01 M and 0.001 M KCl solutions in the same manner.
8. Note the standard specific conductance values of the KCl solutions at the experimental temperature from standard tables (or literature values).

9. Calculate the cell constant for each solution using the formula: Cell Constant = Specific Conductance / Observed Conductance.
10. Take the mean of the cell constant values obtained from different concentrations.

OBSERVATIONS / TABULAR FORM

Temperature of the laboratory / solutions = _____ °C

Conductance of conductivity water (if measured) = _____ S (or mS)

Table 1: Determination of Cell Constant

S. No.	**Concentration of KCl**	**Observed Conductance (G) (S or mS)**	**Specific Conductance (κ) (S cm^{-1})**	**Cell Constant (K = κ / G) (cm^{-1})**
1	0.1 M		0.01288*	
2	0.01 M		0.001413*	
3	0.001 M		0.000147*	
	Mean Cell Constant			

* Approximate specific conductance values of KCl solutions at 25 °C. Use the exact literature values corresponding to the experimental temperature.

Note: If the meter displays conductance in mS, convert to S (1 mS = 10^{-3} S) before calculation, or keep units consistent throughout.

CALCULATION

Cell Constant (K) = Specific Conductance (κ) / Observed Conductance (G)

Example (illustrative):

For 0.01 M KCl at 25 °C,

Specific conductance (κ) = 0.001413 S cm^{-1}

Suppose observed conductance (G) = 0.00155 S

Then, Cell Constant = 0.001413 / 0.00155 = 0.9116 cm^{-1}

Similarly calculate for other concentrations and find the mean value.

Mean Cell Constant = $(K_1 + K_2 + K_3) / 3$ = _____ cm^{-1}

RESULTS

The cell constant of the given conductivity cell at _____ °C is found to be _____ cm^{-1}.

PRECAUTIONS

- Use only conductivity water (very low conductivity) for preparing solutions and rinsing.
- The conductivity cell must be thoroughly rinsed with the solution being measured before taking the reading.
- Ensure that the electrodes are completely immersed and free from air bubbles.
- Note the temperature accurately, as specific conductance varies with temperature.
- Do not touch the platinum electrodes with bare hands.
- Allow the meter reading to stabilize before recording.
- Prepare solutions accurately using analytical balance and volumetric glassware.

VIVA QUESTIONS

1. What is cell constant?

Cell constant is the ratio of the distance between the electrodes (l) to the area of cross-section of the electrodes (a). It is denoted by K or G* and has the unit cm^{-1}.

2. Why is cell constant determined experimentally instead of measuring l and a directly?

Due to edge effects, imperfect geometry of electrodes and the difficulty in measuring the exact effective area and distance, the cell constant is determined experimentally using a standard solution of known specific conductance.

3. Write the formula for cell constant.

Cell Constant = Specific Conductance (κ) / Observed Conductance (G) = $\kappa \times R$ (where R is resistance).

4. Why is KCl solution used for determining cell constant?

The specific conductance of KCl solutions of different concentrations at various temperatures is accurately known and tabulated in literature. KCl is also highly pure and stable.

5. What is the unit of cell constant?

cm^{-1} or m^{-1}.

6. What is the relation between specific conductance and cell constant?

κ = Cell Constant × Observed Conductance.

7. Why is conductivity water used in this experiment?

Ordinary distilled water has appreciable conductivity due to dissolved CO_2 and ions. Conductivity water has extremely low conductivity and does not interfere with the measurements.

8. Does cell constant depend on the concentration of the electrolyte?

Ideally, cell constant is independent of concentration. It depends only on the geometry of the cell. Minor variations may occur due to experimental errors.

9. What is the difference between conductance and specific conductance?

Conductance (G) is the reciprocal of resistance of the solution between the electrodes. Specific conductance (κ) is the conductance of a solution of unit length and unit cross-sectional area. κ = G × cell constant.

10. Why is alternating current (AC) used in conductivity measurements?

Direct current (DC) causes electrolysis and polarization at the electrodes, which alters the resistance. AC prevents these effects and gives accurate conductance values.

11. What is the effect of temperature on specific conductance?

Specific conductance increases with increase in temperature (approximately 2% per degree Celsius) because ionic mobility increases.

12. How is the cell constant used in further experiments?

Once determined, the cell constant is used to calculate the specific conductance (and hence molar or equivalent conductance) of any unknown electrolyte solution by measuring its observed conductance.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT – 12

CHARGING AND DISCHARGING OF A LITHIUM-ION BATTERY — SIMULATION PROCEDURE

Aim

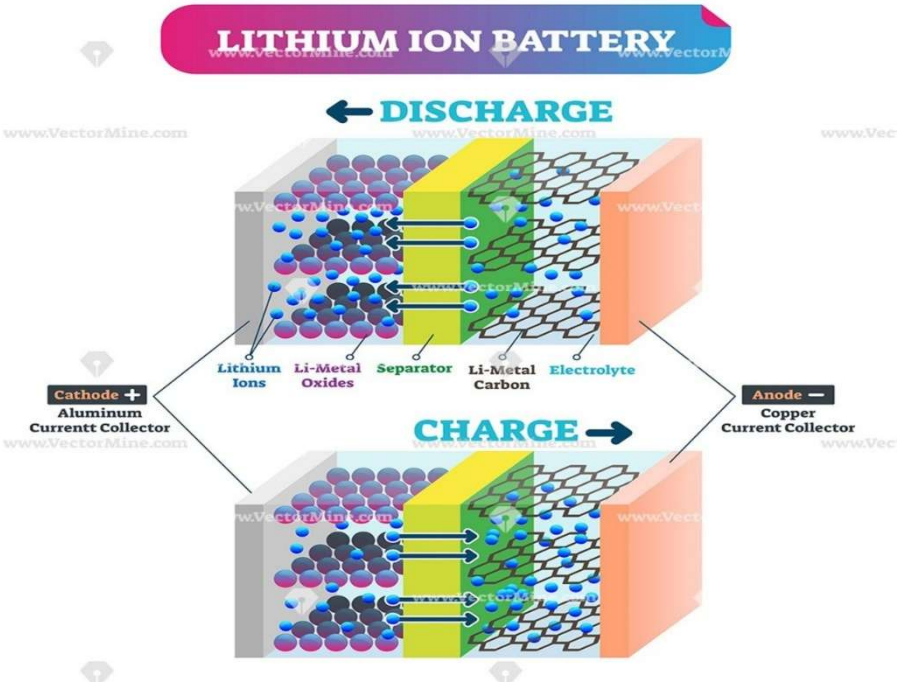
To study the **charging and discharging characteristics of a lithium-ion battery** using a simulation and to determine the variation of **voltage, current, and state of charge (SOC)** with time.

Principle

A lithium-ion battery consists mainly of a **graphite anode, lithium-metal-oxide cathode, and electrolyte**.

- **During discharging:** Lithium ions move from the anode to the cathode, while electrons flow through the external circuit to the load.
- **During charging:** Lithium ions move from the cathode to the anode through the electrolyte, while electrons flow through the external circuit toward the anode.

The battery voltage and current are monitored as functions of time.



Chemical Equations:

During discharge, lithium is oxidized from Li to Li+ (0 to +1 oxidation state) in the lithium-graphite anode through the following reaction:

$$\mathbf{C_6Li \rightarrow\ 6C(graphite) + Li^+ + e^-}$$

These lithium ions migrate through the electrolyte medium to the cathode, where they are incorporated into lithium cobalt oxide through the following reaction, which reduces cobalt from a +4 to a +3 oxidation state:

$$\mathbf{CoO_2\ (s) + Li^+ + e^- \rightarrow\ LiCoO_2\ (s)}$$

Here is the full reaction (left to right = discharging, right to left = charging):

$$\mathbf{C_6Li + CoO_2 \rightleftarrows C_6 + LiCoO_2}$$

These reactions can be run in reverse to recharge the cell. In this case, the lithium ions leave the lithium cobalt oxide cathode and migrate back to the anode, where they are reduced back to neutral lithium and reincorporated into the graphite network.

Simulation Procedure

1. Open the **Lithium-Ion Battery charging/discharging simulation** in the laboratory simulation software.
2. Select the **Lithium-Ion Battery** model.
3. Connect the battery to a **DC power source/charger** through the charging circuit.
4. Set the required battery parameters, such as:
    - Initial state of charge (SOC)
    - Battery capacity
    - Charging voltage
    - Charging current
    - Cut-off voltage
5. Initially keep the battery at a low or predetermined SOC.
6. Start the **charging simulation**.
7. Observe and record the **battery voltage, charging current, SOC, and time** at regular intervals.
8. Continue charging until the battery reaches the specified **upper cut-off voltage/SOC**.
9. Stop the charger and allow the battery to reach the required initial condition for the discharge experiment, if necessary.

10. Disconnect the charger and connect the battery to a **resistive/electronic load**.
11. Start the **discharging simulation**.
12. Record the **battery voltage, discharge current, SOC, and time** at regular intervals.
13. Continue discharging until the battery reaches the specified **lower cut-off voltage**.
14. Stop the simulation and tabulate the observations.
15. Plot: 3 Graphs such as from the observation table

- **Voltage vs. Time**
- **Current vs. Time**
- **SOC vs. Time**



16. Compare the charging and discharging curves and note the changes in battery voltage and SOC.

Observation Table

S.No	Time (s)	Voltage (V)	Current (A)	Capacity (Ah)	SOC (%)	Phase
1	0	3.228733	1.5	0	15	Charging
2	300	3.272195	1.5	0.125	20.68182	Charging
3	600	3.315497	1.5	0.25	26.36364	Charging
4	900	3.358709	1.5	0.375	32.04545	Charging
5	1200	3.401866	1.5	0.5	37.72727	Charging
6	1500	3.444992	1.5	0.625	43.40909	Charging

7	1800	3.488103	1.5	0.75	49.09091	Charging
8	2100	3.531211	1.5	0.875	54.77273	Charging
9	2400	3.574331	1.5	1	60.45455	Charging
10	2700	3.617476	1.5	1.125	66.13636	Charging
11	3000	3.660668	1.5	1.25	71.81818	Charging
12	3300	3.703936	1.5	1.375	77.5	Charging
13	3600	3.747335	1.5	1.5	83.18182	Charging
14	3900	3.790987	1.5	1.625	88.86364	Charging
15	4200	3.835285	1.5	1.75	94.54545	Charging
16	4500	3.601089	-2	0.005556	99.74747	Discharging
17	4800	3.536643	-2	0.172222	92.17172	Discharging
18	5100	3.478168	-2	0.338889	84.59596	Discharging
19	5400	3.420278	-2	0.505556	77.0202	Discharging
20	5700	3.362616	-2	0.672222	69.44444	Discharging
21	6000	3.305066	-2	0.838889	61.86869	Discharging
22	6300	3.247571	-2	1.005556	54.29293	Discharging
23	6600	3.190093	-2	1.172222	46.71717	Discharging
24	6900	3.132602	-2	1.338889	39.14141	Discharging
25	7200	3.075062	-2	1.505556	31.56566	Discharging
26	7500	3.01742	-2	1.672222	23.9899	Discharging
27	7800	2.959571	-2	1.838889	16.41414	Discharging
28	8100	2.901221	-2	2.005556	8.838384	Discharging
29	8400	2.840004	-2	2.172222	1.262626	Discharging
30	8700	2.815	-2	2.2	0	Discharging
31	9000	2.815	-2	2.2	0	Discharging

Expected Result

- During **charging**, the SOC increases and the battery voltage generally increases.
- During **discharging**, the SOC decreases and the battery voltage gradually decreases.
- The charging and discharging curves demonstrate the characteristic behavior of a lithium-ion battery.
- The **charge–discharge efficiency** can be estimated by comparing the energy supplied during charging with the energy delivered during discharging.

Result: The charging and discharging characteristics of a lithium-ion battery were successfully studied using simulation and the following points were explored from the graphs.

1.

2.

3.

VIVA QUESTIONS

1. **What is the aim of the experiment?**

To simulate and study the **charging and discharging characteristics of a lithium-ion battery** by observing voltage, current, capacity, state of charge (SOC), and time.

2. **What is a lithium-ion battery?**

A rechargeable battery in which **lithium ions move between the cathode and anode** during charging and discharging.

3. What happens during charging?

Electrical energy is supplied to the battery, and lithium ions move from the **cathode to the anode**.

4.**What happens during discharging?**

Lithium ions move from the **anode to the cathode**, while electrons flow through the external circuit and deliver electrical energy.

5. **What is SOC?**

SOC means **State of Charge**. It represents the percentage of charge remaining in the battery.

6. **What is the unit of battery capacity?**

Battery capacity is generally expressed in **ampere-hours (Ah)**.

7. **What is the meaning of 1 Ah?**

A capacity of 1 Ah means the battery can theoretically supply **1 ampere for 1 hour** under specified conditions.

8. **What is the difference between charging current and discharging current?**

Charging current is supplied **into the battery**, whereas discharging current flows **out of the battery** to the load.

9. **Why is the current shown as negative during discharge in the simulation?**

The negative sign indicates that current is flowing **out of the battery**, according to the chosen sign convention.

10. **What is the nominal voltage of a typical Li-ion cell?**

Approximately **3.6–3.7 V**, although it varies with the battery chemistry.

11. **What is the typical maximum charging voltage of a conventional Li-ion cell?**

Approximately **4.2 V per cell** for many conventional Li-ion chemistries.

12. **What is the C-rate?**

C-rate indicates the charging or discharging current relative to the battery's rated capacity. For example, **1C** means charging or discharging the battery at a current equal to its rated capacity in Ah.

13. **What is meant by 2C discharge?**

A 2C discharge means the battery is discharged at a current equal to **twice its rated capacity per hour**.

14. **Why does battery voltage change during discharge?**

Battery voltage changes because of changes in **SOC, electrode potential, internal resistance, temperature, and electrochemical reactions**.

15. **Why does the voltage initially drop when discharge begins?**

The initial drop is partly due to the battery's **internal resistance and polarization effects**.

16.What is internal resistance?

It is the effective resistance inside the battery that causes a voltage drop when current flows.

17. **What is the relationship between voltage, current, and resistance?**

According to Ohm's law:

V=IRV=IR

18.**What is battery efficiency?**

Battery efficiency is the ratio of energy delivered during discharge to the energy supplied during charging, usually expressed as a percentage.

19. **What is Coulombic efficiency?**

It is the ratio of the charge removed during discharge to the charge supplied during charging:

$$\eta C = \frac{Q_{\text{discharge}}}{Q_{\text{charge}}} \times 100$$

20. **Why is simulation useful for battery studies?**

Simulation allows us to study battery behavior **safely, quickly, and economically** without repeatedly performing physical experiments.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty

EXPERIMENT - 13

DETERMINATION OF OPTICAL ENERGY BAND GAP OF SEMICONDUCTORS FROM ABSORPTION SPECTRA USING SIMULATION

Aim

To study the variation of **absorbance with wavelength** for different semiconductor materials—Zinc Oxide(**ZnO), Gallium Arsenide (GaAs), Silicon(Si), Cadmium Sulphide(CdS) and (Titanium Dioxide) TiO_2**—using simulation and to determine/compare their approximate optical band-gap energies.

Materials selected

1. Zinc Oxide (ZnO)
2. Gallium Arsenide (GaAs)
3. Silicon (Si)
4. Cadmium Sulphide (CdS)
5. Titanium Dioxide (TiO_2)

Principle

When light falls on a semiconductor, photons having energy greater than or equal to the band-gap energy can excite electrons from the **valence band to the conduction band**.

The energy of a photon is: $E=hc/\lambda$

where:

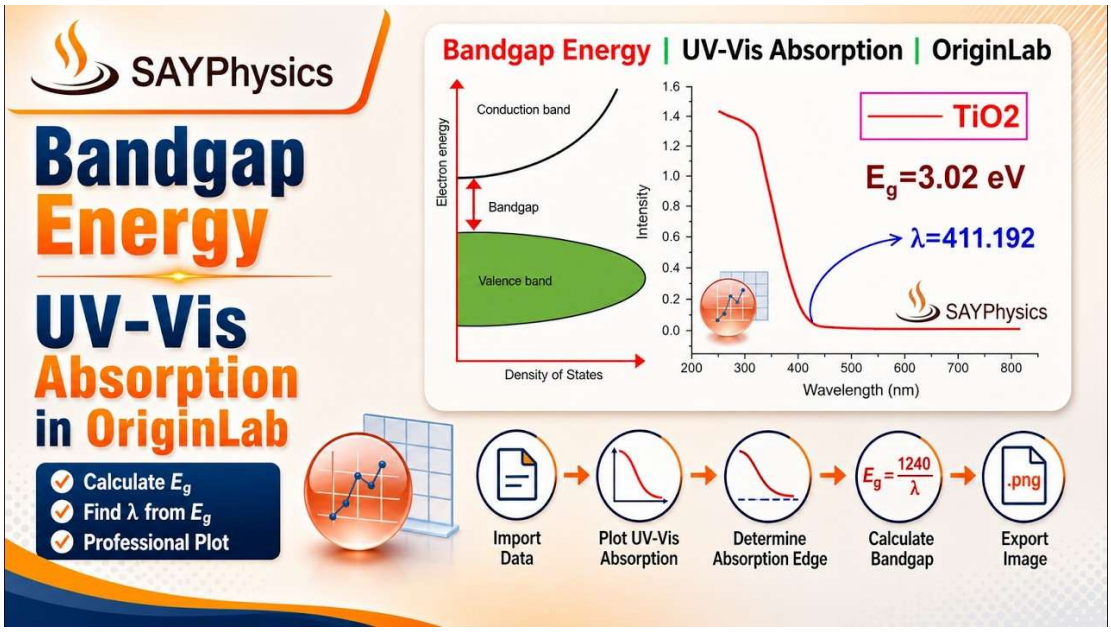
- E = photon energy
- h = Planck's constant
- c = velocity of light
- λ(lambda) = wavelength

For band-gap $Eg=hc/\lambda$

In electron-volt units:

$$Eg(eV)=1240/\lambda g(nm)$$

where λg\lambda g is the **absorption-edge wavelength**.



Absorbance

The absorbance of a material is related to the intensity of incident and transmitted light by:

$A=-\log_{10}(I/I0)$, where:

- I_0 = incident light intensity
- I = transmitted light intensity
- A = absorbance

For a sample obeying Beer–Lambert behavior:

$A=\epsilon cl$

Ex: Absorbance of a 0.8N solution having the molar absorption coefiicient as 0.8 is, $A=\epsilon cl=1.2(0.8)(1)=0.96$ absorbance units.

Simulation Procedure:

Step 1: Open the simulation

Open the semiconductor optical absorption/UV-Visible spectroscopy simulation.

Step 2: Select the semiconductor

Select **ZnO** from the material list.

Step 3: Select wavelength range

Set an appropriate wavelength range, for example: 200 nm − 1000 nm

For wide-band-gap materials such as ZnO and TiO_2, the UV region is particularly important.

Step 4: Record absorbance

Change the wavelength in suitable intervals, for example: **20 nm or 25 nm intervals**, depending on the simulation. Record the corresponding absorbance.

Step 5: Repeat the same procedure for:

- ZnO
- GaAs
- Si
- CdS
- TiO_2

Keep the simulation conditions identical wherever possible.

Observation Table

Wavelength (nm)	**ZnO Absorbance**	**GaAs Absorbance**	**Si Absorbance**	**CdS Absorbance**	**TiO_2 Absorbance**
200					
250					
300					
350					
400					
450					
500					
550					
600					
650					
700					
750					
800					
900					
1000					

Step 6: Plot the graph

Plot a graph of:

Absorbance vs Wavelength

X-axis : Wavelength (nm)

Y-axis : Absorbance (A.U.)

You can plot all five materials on the same graph for comparison.

Step 7: Determine absorption edge

From each absorption spectrum, identify the wavelength corresponding approximately to the **absorption edge λg\lambda_g**.

The absorption edge is the region where the absorbance begins to increase strongly due to electronic transitions across the band gap.

Then calculate:

$E_g = 1240/\lambda_g$

Expected Comparison

Approximate room-temperature values are:

Semiconductor	Approx. EgE_g	Approx. absorption-edge wavelength*
ZnO	3.3–3.4 eV	~365–375 nm
GaAs	1.42 eV	~870 nm
Si	1.12 eV	~1100 nm
CdS	2.4 eV	~515 nm
TiO_2	~3.0–3.2 eV	~390–415 nm

*Approximate values; actual simulated spectra depend on the material model, crystal phase, temperature and simulation parameters.

Approximate relationship

$\lambda_g = 1240/E_g$

Therefore:

Higher band gap⇒shorter absorption-edge wavelength

Lower band gap⇒longer absorption-edge wavelength

For example, ZnO has a large band gap, so its absorption edge occurs in the **UV region**, whereas Si has a smaller band gap and its absorption extends toward the **near-infrared region**.

Expected Graph

The graph should show that the materials begin absorbing strongly at different wavelength regions.

Approximate order of absorption-edge wavelength:

ZnO<TiO2<CdS<GaAs<Si

This is essentially the reverse of the band-gap-energy order Si<GaAs<CdS<TiO2<ZnO.

Calculation Example

Suppose the simulated absorption edge of CdS is:

$\lambda g = 515$ nm

Then:

$Eg = 1240515$

Therefore,

$Eg \approx 2.41$ eV

which is close to the expected band gap of CdS.

Result

The absorbance spectra of **ZnO, GaAs, Si, CdS and TiO_2** are obtained using simulation. The absorption-edge wavelength of each semiconductor is determined, and the corresponding optical band-gap energy is calculated using:

$Eg(eV) = 1240/\lambda g(nm)$

The experiment demonstrates that **semiconductors with larger band gaps absorb shorter-wavelength, higher-energy photons**, while smaller-band-gap semiconductors absorb at longer wavelengths.

Viva Questions

Basic Questions

1. What is the aim of the experiment?

To study the variation of absorbance with wavelength for different semiconductors and determine their approximate optical band-gap energies from the absorption edge.

2. What is absorbance?

Absorbance is a measure of how much light is absorbed by a material.

$$A = -\log I_t/(I_0)$$

where I_0 is incident intensity and I_t is transmitted intensity.

3. What is the band gap of a semiconductor?

It is the minimum energy required to excite an electron from the valence band to the conduction band.

4. What is the relation between photon energy and wavelength?

$E = hc/\lambda$ or, in electron volts, $E(eV) = 1240/\lambda(nm)$

5. What is absorption edge?

The absorption edge is the wavelength region at which strong optical absorption begins due to electronic transitions across the band gap.

6. How do you determine the band gap from the absorption edge?

$Eg = 1240/\lambda g$, where λg is the absorption-edge wavelength in nm.

Questions on the Graph

7. What is plotted on the X-axis?

Wavelength in nanometres (nm).

8. What is plotted on the Y-axis?

Absorbance, usually in arbitrary units (A.U.).

9. Why does absorbance increase sharply near the absorption edge?

Because photons acquire sufficient energy to promote electrons across the band gap, producing strong absorption.

10. What happens when the wavelength of light decreases?

Photon energy increases because: $E \propto 1/\lambda$

11. What happens to the absorption edge when the band gap increases?

The absorption edge shifts toward a **shorter wavelength**.

12. What is the relationship between band gap and absorption wavelength?

$Eg \propto 1/\lambda_g$, Thus, larger band gap means shorter absorption wavelength.

Questions on the Five Semiconductors

13. Which semiconductor among the five has the smallest band gap?

Silicon (Si), approximately **1.12 eV** at room temperature.

14. Which has the largest band gap among the selected materials?

ZnO, approximately **3.3–3.4 eV**.

15. What is the approximate band gap of GaAs?

Approximately **1.42 eV**.

16. What is the approximate band gap of CdS?

Approximately **2.4 eV**.

17. What is the approximate band gap of TiO_2?

Approximately **3.0–3.2 eV**, depending on its crystal phase.

18. Arrange the materials in increasing order of band gap. Si<GaAs<CdS<TiO2<ZnO

19. Arrange them in increasing order of absorption-edge wavelength.

ZnO<TiO2<CdS<GaAs<Si

20. Why does Si absorb at a longer wavelength than ZnO?

Si has a much smaller band gap, so lower-energy, longer-wavelength photons can produce electronic transitions.

Conceptual Viva Questions

21. What is a semiconductor?

A material whose electrical conductivity lies between that of a conductor and an insulator and whose conductivity generally increases with temperature.

22. What are valence and conduction bands?

The valence band contains electrons involved in bonding, while the conduction band contains electrons that can participate in electrical conduction.

23. What happens when a photon has energy less than the band gap?

It generally cannot excite an electron across the fundamental band gap, so strong band-to-band absorption does not occur.

24. What happens when photon energy is equal to or greater than the band gap?

An electron can be excited from the valence band to the conduction band.

25. What is meant by a direct band-gap semiconductor?

A semiconductor in which the conduction-band minimum and valence-band maximum occur at the same crystal momentum.

26. Which of the selected materials is a direct band-gap semiconductor?

GaAs and CdS are commonly treated as direct band-gap semiconductors.

27. Is silicon a direct or indirect band-gap semiconductor?

Silicon is an **indirect band-gap semiconductor**.

28. Why is GaAs important in optoelectronics?

Its direct band gap makes it efficient for light emission and absorption, so it is widely used in optoelectronic devices.

29. Why is ZnO useful in UV applications?

Its wide band gap means it interacts strongly with relatively high-energy UV photons.

30. Why is TiO_2 used in optical and photocatalytic applications?

It has a wide band gap and can absorb UV radiation, enabling photo-induced electronic processes.

EVALUATION SHEET

S. No.	Component	Max . Marks	Marks Secured
1	Preparedness		
2	Viva-voce		
3	Experiment		
4	Analysis		
5	Record		
6	Total		

Signature of the faculty